

# TFPT — $E_8$ Audit, Cascade Bridge and Bootstrap

The seven  $E_8$  slices as an audit raster, the old cascade as the same even-integer spine, the Möbius self-consistency loop — and the thirty-step celestial/twistor route with the measure chain derived

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July 23, 2026 · v5.4 (rev 389)

## What this document is about

$E_8$  as an *audit container*, not a mystery: the seven maximal slices of 248 as a falsification raster (every load-bearing number must appear in at least one projection), the bridge showing the old  $E_8$  orbit cascade  $D=60-2n$  is the same even-integer spine as the compiler, and the Möbius bootstrap in which  $g_{\text{car}} = 5$  and the “8” in  $c_3$  are overdetermined  $E_8$ -closure fixed points (only  $\pi$  irreducible). Since the celestial/twistor round the note also carries the **thirty-step continuum story** (§17) — from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow — in which the **measure chain of the celestial route is derived, not declared**: single-valuedness from F-independence + the Quillen pairing (v520), the  $1/\det_j$  normalisation the Atiyah–Bott fixed-point factor computed from three independent sources with  $\psi = 64$  reproduced (v523), the residual **[O]** narrowed to the global BCOV integral beyond the fibre zero-mode factor; plus the Woit OS-bridge stages  $\alpha/\beta_1/\beta_2$  (v519/v522/v524), the thermal seam (v526), the ten-test side-blind scoreboard with the twist-class definition of the bit (v528) and the first firing interacting kill test under its typed fence (v529). SEAM.EQUIV.01 and WOIT.OS.TWISTOR.01 stay **[O]**; no marker moves.

## What $E_8$ is *not* (and why the known no-go results do not apply)

$E_8$  here is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures — **not** an unbroken physical gauge group. The Standard Model is a *readout* after projection (not a direct  $E_8$  gauge theory), the Lorentz signature appears only after projection, and fermions are not “everything in the adjoint”. Consequently the standard objections to literal  $E_8$  world-formulas do *not* bite: Distler–Garibaldi (no chiral fermions / representation obstructions for  $E_8$  as a 4d gauge group) and Coleman–Mandula (no nontrivial mixing of spacetime and internal symmetry) both constrain  $E_8$  as a *spacetime gauge symmetry*, which TFPT does not claim. The defensible statement is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull* — a discrete classifier, attacked on its own (combinatorial) terms, not on gauge-theoretic ones.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** R. **tfpt\_research\_contracts** — the open interfaces  $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$  as numbered

contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S. tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

**You are reading document 3 ( $E_8$  Audit & Bootstrap).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

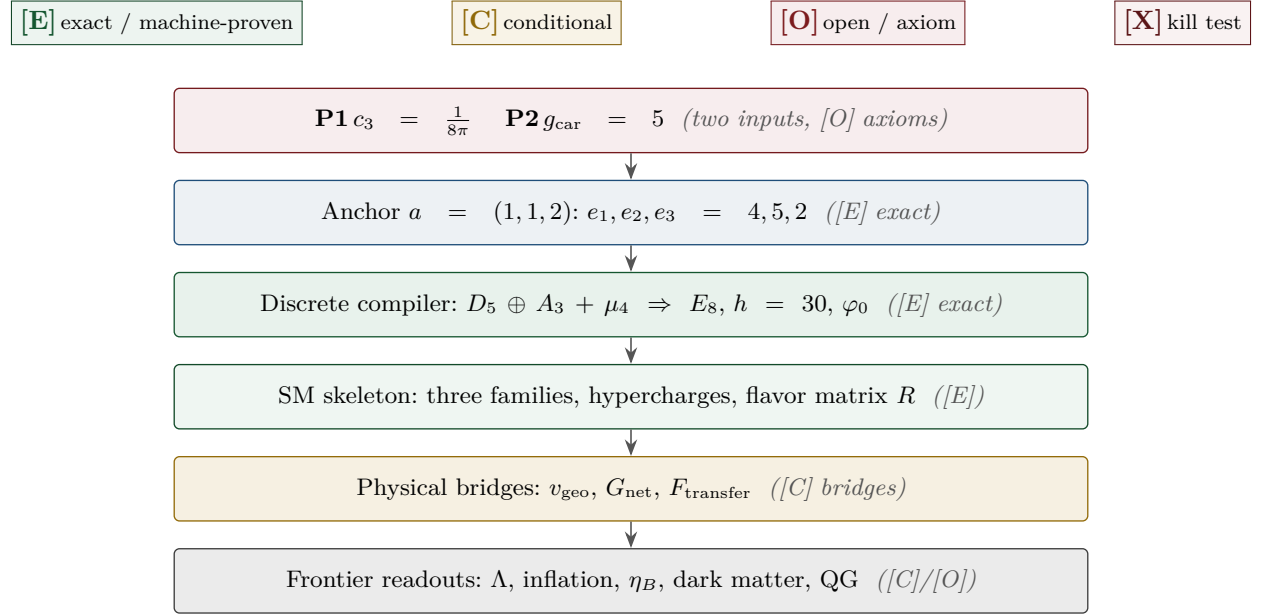
**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the **#print axioms** check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route **seamResidualClosed'** (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of **verification/status\_ledger.csv** (the single source of truth); **if the text and the ledger ever disagree, the ledger wins**. The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in **verification/status\_ledger.csv**.



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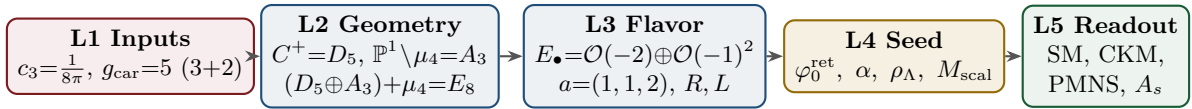
## Part I

# The $E_8$ secondary-branching atlas

**Purpose and discipline.** This note does *not* introduce new physics. It builds an *audit raster*: the claim that *every load-bearing TFPT number should appear in at least one  $E_8$  branching projection*. A number that appears in none is suspect; a number appearing in several independent projections becomes structurally strong. All arithmetic identities below are exact [E]; the *physical readings* are audit-level [O] unless tied to a derivation. (All 77 numeric claims were machine-checked with exact integer/rational arithmetic.)

## 1 The compiler core (where this note sits)

The strongest current form of the theory is a five-layer compiler:



$E_8$  is now a *closure* object, not an input: the carrier is the  $D_5 = \mathfrak{so}_{10}$  half-spinor, the four punctures  $\mathbb{P}^1 \setminus \mu_4$  give  $A_3 = \mathfrak{su}_4$ , and the common discriminant group  $\mathbb{Z}_4$  glues them, with the norm test  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ . This note charts the *secondary* maximal subalgebras of  $E_8$  as audit windows onto the TFPT modules.

## 2 The (1, 1, 2) anchor microcode

The parabolic anchor  $a = (1, 1, 2)$  (splitting type of  $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ) is a miniature compiler. Its power sums  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the discrete budget:

| $n$ | $p_n = 2 + 2^n$ | TFPT reading (verified)                                                           |
|-----|-----------------|-----------------------------------------------------------------------------------|
| 1   | 4               | $ \mu_4  = -\deg E_\bullet$ (glue index)                                          |
| 2   | 6               | $ R^+(A_3) $ (positive $A_3$ roots) = $\mathbb{Z}_6$ cusp order                   |
| 3   | 10              | $\mathcal{A}_\Lambda = \binom{5}{2} = \dim V_{D_5}$ (action ladder / pair sector) |
| 4   | 18              | $N_{\text{fam}}  R^+(A_3)  = 3 \cdot 6$                                           |
| 5   | 34              | $2h(D_5) + 3 R^+(A_3)  = 16 + 18$ ; and $34 -  R(A_3)  = 22 = \sum R$             |

and the budget compresses to products of power sums:

$$q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}, \quad \sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1 p_2 = 48, \quad D_{\text{start}} = p_2 p_3 = 60, \quad |R(E_8)| = 4p_2 p_3 = 240.$$

The first three power sums alone reach  $E_8$ :  $p_1 p_2 p_3 = 4 \cdot 6 \cdot 10 = 240 = |R(E_8)|$ , and  $p_1 p_2 p_3 + (p_4 - p_3) = 240 + 8 = 248 = \dim E_8$ .

### The anchor difference ladder: the binary carrier spine [E]

While the *products* of the power sums give the large budgets, the *differences* give the binary carrier spine in one line:

$$p_n(a) = 2 + 2^n \implies \Delta p_n = p_{n+1} - p_n = 2^n : \quad (2, 4, 8, 16) = (|\mathbb{Z}_2|, |\mu_4|, h(D_5), \dim S^+).$$

So the single anchor  $a = (1, 1, 2)$  generates the big numbers multiplicatively and the doubling spine 2, 4, 8, 16 additively — one vector, two readings.

### The anchor reconstructs the carrier rank — and the 41-chain

The characteristic polynomial of the anchor vector is  $(t-1)^2(t-2) = t^3 - 4t^2 + 5t - 2$ , with elementary symmetric data

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|_{\text{sheet}}.$$

So the parabolic flavor geometry reads the carrier rank  $g_{\text{car}} = 5$  *backwards*:  $D_5 \Rightarrow a$  but also  $a \Rightarrow g_{\text{car}}$ . Moreover  $3^2 + 4^2 = 5^2$  sits as  $(p_0, e_1, e_2) = (3, 4, 5)$ , giving the *anchor form* of the abelian index

$$10b_1 = e_1(a)^2 + e_2(a)^2 = 4^2 + 5^2 = 41$$

alongside the known  $41 = 40 + 1 = \sum L + N_{\Phi}$ . [E]

### Pascal closure: $g_{\text{car}} = 5$ is the unique solution (kills alternatives) [E]

The carrier rank is not merely a good fit — it is the *unique* root of an exact closure. The even-exterior carrier dimension  $2^{g-1}$  must equal the truncated Pascal sum  $\binom{g}{0} + \binom{g}{1} + \binom{g}{2}$  (the  $1+g+\binom{g}{2}$  that builds  $\dim S^+$ ):

$$2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2} \iff g = 5$$

( $g=3 : 4 \neq 7$ ;  $g=4 : 8 \neq 11$ ;  **$g=5 : 16=16$** ;  $g=6 : 32 \neq 22$ ;  $g=7 : 64 \neq 29$ ). At  $g = 5$  the row is  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ , the same Pascal triple that reads family, action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . This is the fourth independent lock on  $g_{\text{car}} = 5$  (with rank-fill, Coxeter-match, and integer-glue/norm), and unlike those it explicitly *kills* the neighbours  $g = 4, 6$ .

**The Pascal Ladder Theorem** ([[verification/v108\\_pascal\\_ladder.py](#)]). The closure generalises exactly: for *every* truncation degree  $K$ ,

$$2^{g-1} = \sum_{k \leq K} \binom{g}{k} \iff g = 2K + 1$$

(odd  $g$  solves by the binomial midpoint identity; even  $g$  is excluded by the strict straddle  $\sum_{k < g/2} < 2^{g-1} < \sum_{k \leq g/2}$ ). **Consequence (zero slack): the carrier selection  $g = 5$  is *equivalent* to the truncation degree  $K = 2$**  — the Pascal-selection residual of the red team (Target B) is therefore exactly the one named hypothesis *Quadratic Boundary Locality* (the seam certifies code degrees  $\leq 2$ ; the architecture document). Neighbour worlds on the ladder:  $K=1 \rightarrow g=3$  (a *one*-family world),  $K=2 \rightarrow g=5$  (three families),  $K=3 \rightarrow g=7$  (*nine* families),  $K=4 \rightarrow g=9$  *inconsistent* ( $N_{\text{fam}} = 255/9 \notin \mathbb{Z}$ ). Overdetermination: among the ladder worlds, only  $K = 2$  also satisfies the independent rank closure  $g + N_{\text{fam}} = 8 = \text{rank } E_8$  — two independent selections agree on the same world. The hypothesis itself stays [C].

### The seed is an anchor expression

The retained seed is exactly the two-term anchor combination of  $c_3$ :

$$\varphi_0^{\text{ret}} = \frac{2p_1}{p_2} c_3 + 2p_1 p_2 c_3^4 = \frac{4}{3} c_3 + 48 c_3^4 = \frac{1}{6\pi} + \frac{3}{256\pi^4}$$

(verified to 30 digits). The Cabibbo base is  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , the hierarchy engine of the whole fermion spectrum. [E]

## 3 The residue and length matrices as spectral objects

With the residue matrix  $R$  and the full word-length matrix  $L = R + 6W$  ( $W$  the rank-one winding),

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \quad L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix}, \quad W = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

the spectral invariants are pure compiler numbers (all verified):

| Invariant of $R$      | value                  | Invariant of $L$      | value                                  |
|-----------------------|------------------------|-----------------------|----------------------------------------|
| $\text{tr } R$        | $9 = N_{\text{fam}}^2$ | $\text{tr } L$        | 15                                     |
| $\det R$              | $8 = h(D_5)$           | $\det L$              | $20 =  R(A_4)  = 2\mathcal{A}_\Lambda$ |
| $\chi_R(t)$           | $t^3 - 9t^2 + 10t - 8$ | $\chi_L(t)$           | $t^3 - 15t^2 + 40t - 20$               |
| $\text{PrinMin}_2(R)$ | $(2, 3, 5)$            | $\text{PrinMin}_2(L)$ | $(5, 14, 21)$                          |
| $\ R\ _F^2$           | $78 = \dim E_6$        | —                     | —                                      |

The three principal  $2 \times 2$  minors of  $R$  are  $(2, 3, 5)$  — sheet, family, carrier — with product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  and sum  $10 = \mathcal{A}_\Lambda$ . The wrong down-branch  $\{1, 3, 4\}$  fails exactly these invariants.

**Bilinear anchor forms.** With  $\mathbf{1} = (1, 1, 1)^\top$  and  $a = (1, 1, 2)^\top$  (all verified):

| $R$               | $\mathbf{1}$  | $a$             | $L$               | $\mathbf{1}$                  | $a$                           |
|-------------------|---------------|-----------------|-------------------|-------------------------------|-------------------------------|
| $\mathbf{1}^\top$ | $22 = \sum R$ | $27 = 3^3$      | $\mathbf{1}^\top$ | $40 =  R(D_5) $               | $45 = \dim D_5$               |
| $a^\top$          | $32 = 2^5$    | $40 =  R(D_5) $ | $a^\top$          | $56 = \dim \mathbf{56}_{E_7}$ | $64 = \dim S^+ \cdot  \mu_4 $ |

The corner value  $a^\top L a = 64 = (16, 4)$  reads exactly one spinor-gluon sheet of the  $E_8$  branching. And the rank-one winding update shows up in the inverses:

$$R^{-1} \mathbf{1} = \frac{1}{4}(1, 1, -1)^\top, \quad L^{-1} \mathbf{1} = \frac{1}{10}(1, 1, -1)^\top$$

i.e. the residue response reads the glue denominator  $|\mu_4| = 4$ , the full-length response reads the decouple denominator  $\mathcal{A}_\Lambda = 10$ . [E]

### Anchor-plane Plücker coordinates: scattered hits become oriented areas [E]

The strongest anti-numerology tool is to read *orientation invariants* of the anchor plane  $\langle \mathbf{1}, a \rangle$  rather than loose integers. For  $M \in \{R, Q, K, L\}$  the left block  $B_M = (\mathbf{1}^\top M; a^\top M)$  (a  $2 \times 3$ ) and the right



block  $C_M = (M \mathbf{1} \ Ma)$  (a  $3 \times 2$ ) have the  $2 \times 2$  Plücker minors

|     | Pl( $\cdot$ ) (left)                          | Pl $_R(\cdot)$ (right)                                               |
|-----|-----------------------------------------------|----------------------------------------------------------------------|
| $R$ | $(-6, 2, 14)$                                 | $(8, 12, 4) = (h(D_5),  R(A_3) ,  \mu_4 )$                           |
| $Q$ | $(3, 6, 3)$                                   | $(0, 4, 5) = (0,  \mu_4 , g_{\text{car}})$                           |
| $K$ | $(-1, 6, 4) = (-N_\Phi,  R^+(A_3) ,  \mu_4 )$ | $(12, 12, -2) = ( R(A_3) ,  R(A_3) , - \mathbb{Z}_2 )$               |
| $L$ | $(6, 26, 14)$                                 | $(20, 30, 10) = (2\mathcal{A}_\Lambda, h(E_8), \mathcal{A}_\Lambda)$ |

Three load-bearing flavor numbers are now single oriented areas, not coincidences:

$$\| \text{Pl}(K) \|_1 = 11, \quad \| \text{Pl}_R(K) \|_1 = \text{Pl}(L)_{13} = 26, \quad \sum \text{Pl}_R(L) = 60 = D_{\text{start}}$$

(11 is the mass-power anchor-plane norm; 26 the  $c_t/c_b$  denominator; 60 the  $E_8$  cascade start). The left-null-space response complements the right one above: the column sums of  $\det L \cdot L^{-1}$  and  $\det R \cdot R^{-1}$  are  $(-5, 1, 6)$  and  $(1, -5, 6)$ , the same magnitudes  $\{N_\Phi, g_{\text{car}}, |R^+(A_3)|\} = \{1, 5, 6\}$ . Finally the sector rows themselves are anchor-shifted:  $K_e - K_u = (1, 1, 2) = a$  and  $L_e - L_u = (1, 2, 3) = \text{Exp}(A_3)$  — leptons are up-quarks displaced by the family roots. All entries are exact [\[E\]](#); this is an audit layer, not a closure (the physical quark amplitudes stay [\[C\]](#)). [\[verification/v37\\_plucker\\_anchor.py\]](#)

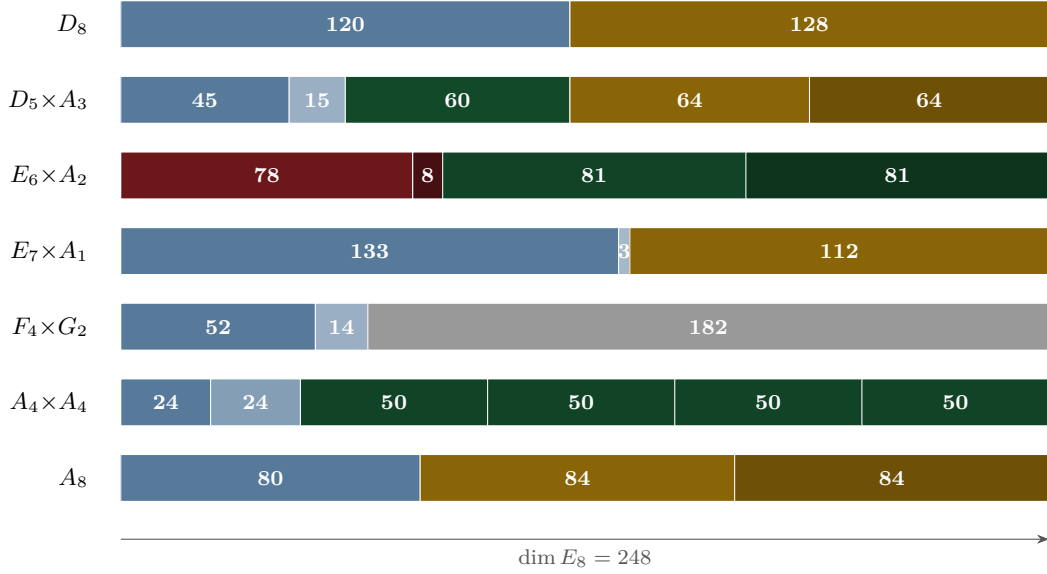


Figure 1: The  $E_8$  slice atlas: seven maximal-subalgebra cuts of 248. Gold = spinor-glue ( $64=(16, 4)$ ,  $112=7 \cdot 16$ ,  $84=7 \cdot 12$ ,  $128$ ); red = flavor-residue ( $78=\|R\|_F^2$ ,  $8=\det R$ ); green = family/action blocks; blue = Lie adjoints.

### The numerology null test: the look-elsewhere burden quantified [E]

The raster above disciplines *which* integers may appear; the complementary statistical question — *could a random “theory” of equal formula complexity land this many data hits?* — is answered by an explicit, fully declared null model. For each of the 13 scored frozen-registry observables the *entire* grammar class is enumerated exactly (no sampling): one-term expressions  $\frac{p}{q} (\varphi_0^{\text{ret}})^a \lambda_C^c (1+\lambda_C)^d \pi^b e^{u/w}$  with  $p, q \leq 130$ ,  $|a| \leq 2$ ,  $|c| \leq 3$ ,  $|d|, |b| \leq 1$ ,  $|u| \leq 6$ ,  $w \leq 6$ , and two-term sums with coefficients  $\leq 12$  — a class that provably contains every TFPT formula it scores (machine-checked fairness condition). Data windows are conservative: the *maximum* of the achieved deviation, the documented experimental  $\sigma$ , the quote resolution and the 5% quark scheme spread, so the known tension observables ( $\sin^2 \theta_{13}$ ,  $s_{23}$ ,  $\delta_{\text{CKM}}$ ,  $m_\mu/m_\tau$ ,  $\beta$ ) enter with their honest *large* windows and contribute almost nothing. Result of the census: joint formula-fishing probability  $\prod_i p_i = 10^{-25.8}$ , stable to  $< 2$  orders of magnitude when the coefficient bound is varied  $80 \dots 200$  and the plausibility window  $5 \times \dots 20 \times$ . Monte-Carlo cross-check (deterministic seeds):  $2 \times 2 \cdot 10^5$  pseudo-theories — random rational dials inside TFPT’s own formula skeletons, with the seed  $\varphi_0^{\text{ret}}$  fixed resp. drawn log-uniformly — reach at most 5/13 resp. 4/13 hits versus TFPT’s 13/13. The test has *power*: perturbing  $\varphi_0^{\text{ret}}$  by 1%/10%/30% collapses TFPT’s own scorecard to 9/6/4 hits, and shuffling which measured value is assigned to which formula collapses it to a mean of 1.2. Separately, all 94500 complexity-matched variants of the  $F_{U(1)}$  fixed-point equation (coefficients,  $M$ , transport exponent, resolvent power) are root-solved: exactly *one* — the TFPT instance  $(2, \frac{4}{5}, M=41, e^{-2\alpha}, -\frac{5}{4})$  — lands inside the achieved  $4 \cdot 10^{-8}$  CODATA window, so  $p_\alpha \leq 1.1 \cdot 10^{-5}$ . Headline:

$$P(\text{null reproduces the scorecard incl. } \alpha) \leq 10^{-30.7} \quad (\approx 102 \text{ bits})$$

*conditional on the declared grammar* — a look-elsewhere-corrected null-model rejection, never “certainty”. Complementary to the frozen registry: the freeze (v84) prevents *future* look-elsewhere drift, this test quantifies the *retrospective* look-elsewhere burden. [verification/v100\_numerology\_null\_mc.py]

## 4 The secondary branching atlas

Each maximal subalgebra of  $E_8$  mirrors a TFPT module. The dimension identities are exact [E]; the readings are audit-level [O]. The seven *slices* of  $\dim E_8 = 248$  are shown below: the same total cut seven ways, each cut a different TFPT module.

| Subalgebra       | 248 =                  | TFPT reading                                                                                                          |
|------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $D_8$            | 120 + 128              | family-transition adjoint + two spinor-gluon sheets<br>( $128=2 \dim S^+ \mu_4 $ )                                    |
| $D_5 \times A_3$ | 45 + 15 + 60 + 64 + 64 | <b>main split:</b> $\dim D_5, \dim A_3, \mathcal{A}_\Lambda  R^+(A_3) $ , two (16, 4) glue sheets                     |
| $E_6 \times A_2$ | 78 + 8 + 81 + 81       | <b>flavor residue:</b> $\ R\ _F^2, \det R = \dim A_2$ , cubic family blocks                                           |
| $E_7 \times A_1$ | 133 + 3 + 112          | <b>scalaron:</b> $112=7 \cdot 16$ (exponent $\times$ one family)                                                      |
| $F_4 \times G_2$ | 52 + 14 + 182          | <b>occupancy/gauge:</b> $ R(F_4) =48=\Omega_{\text{adm}},  R(G_2) =12=\dim \mathfrak{g}_{\text{SM}}$                  |
| $A_4 \times A_4$ | 24 + 24 + 4 · 50       | <b>action ladder:</b> $24+24=48=\Omega_{\text{adm}}$ ; off-diagonal (5, 10) reps $\mathcal{A}_H, \mathcal{A}_\Lambda$ |
| $A_8$            | 80 + 84 + 84           | rank-decuplet + scalaron: $80=8 \cdot 10, 84=7 \cdot 12$                                                              |

| The $D_8$ split is a magnitude/phase channel [E]/[C]/[O]<br>( [verification/v227_degree_exponent_channel_split.py] )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The <math>D_8</math> row <math>248 = 120 + 128</math> is the canonical <i>exponent/degree</i> split of <math>E_8</math>: the exponents sum to <math>120 =  R^+(E_8) </math> (a <b>magnitude</b> channel — positive roots, root transitions, the dimensionless ratio/product readouts), the fundamental degrees <math>\{2, 8, 12, 14, 18, 20, 24, 30\}</math> sum to <math>128 = 2^{\text{rank}-1} = \text{rank } E_8 \cdot \dim S^+</math> (a <b>phase/gluon</b> channel — the spinor/Ramond sheets). This gives the red-team Target D CP residual a clean home: the (ratios, product) <math>\Rightarrow v_{\text{geo}}</math> magnitude bijection lives in the 120 channel, the un-covered CP phases in the 128 channel (cf. the hexagonal phase fiber, red team Target D). Whether 128 also hosts a dark sector is Frontier [O] — this is the honest replacement for the over-strong “<math>S^-</math> is dark matter” reading, not an assertion of it. The conjugate sheet <math>S^-</math> and the double-cover deck sit in this <i>matched</i> channel (<math> \mathbb{Z}_2  = 2</math> and <math>128 = \sum \deg</math>), forced-disjoint from the five unmapped degrees <math>\{12, 14, 18, 20, 24\}</math> (v430): the seam’s “other side” is not the hull overhead.</p> |

| $E_8$ Slice Compression Lemma [E] (audit)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The seven slices are not seven separate narratives — they are <i>one</i> projection of two small alphabets. The anchor power sums <math>p_n(a) = 2 + 2^n</math> give <math>P = (p_0, p_1, p_2, p_3) = (3, 4, 6, 10)</math>, whose six <math>K_4</math> edge products are exactly the carrier blocks <math>(p_0 p_1, \dots, p_2 p_3) = (12, 18, 30, 24, 40, 60) = ( R(A_3) , N_{\text{fam}} R^+(A_3) , h(E_8),  W(A_3) ,  R(D_5) , D_{\text{start}})</math>; the six Sheet-Diamond operators carry determinants <math>(\det Q, \det K, \det R, \det C, \det L, \det F) = (3, 4, 8, 14, 20, 32)</math> with <math>\sum \det = 81 = N_{\text{fam}}^4</math>, the total determinant charge of the flavour space. All seven 248-slices, the row-budget cross <math>\text{row}(C + sU + tV) = (7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)</math> (democratic winding + <math>A_3</math> exponents), and <math>\dim E_6 = 78 = p_2(p_0 + p_3)</math> then follow from <math>P</math>, those determinants and <math>\Delta = p_0 + p_3 = 13</math>. The atlas is therefore a closed grammar over admissible invariant classes (power sums, determinants, branching dimensions), not a shelf of trophies ([verification/v170_diamond_slice_compression.py]).</p> |

**Theorem 1** ( $E_6 \times A_2$  flavor-residue shadow [O]). The principal flavor branching reads the residue matrix:

$$\|R\|_F^2 = 78 = \dim E_6, \quad \det R = 8 = \dim A_2 = h(D_5), \quad \mathbf{1}^\top R a = 27,$$

and the full  $E_6 \times A_2$  dimension count is the residue identity

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3.$$

Here  $E_6$  reads the Frobenius norm of the residue matrix,  $A_2$  the pure three-family symmetry, and the **27** the cubic family block. (Strongest new  $E_8$  flavor fingerprint.)

**Theorem 2** ( $E_7 \times A_1$  scalaron shadow [O]). With the scalaron exponent  $7 = \Omega_{\text{adm}} - 10b_1 = -\deg E + \text{rk } E$ ,

$$248 = 133 + 3 + 112, \quad 112 = 7 \cdot \dim S^+ = 7 \cdot 16, \quad |R(E_7)| = 126 = 7 \cdot 18,$$

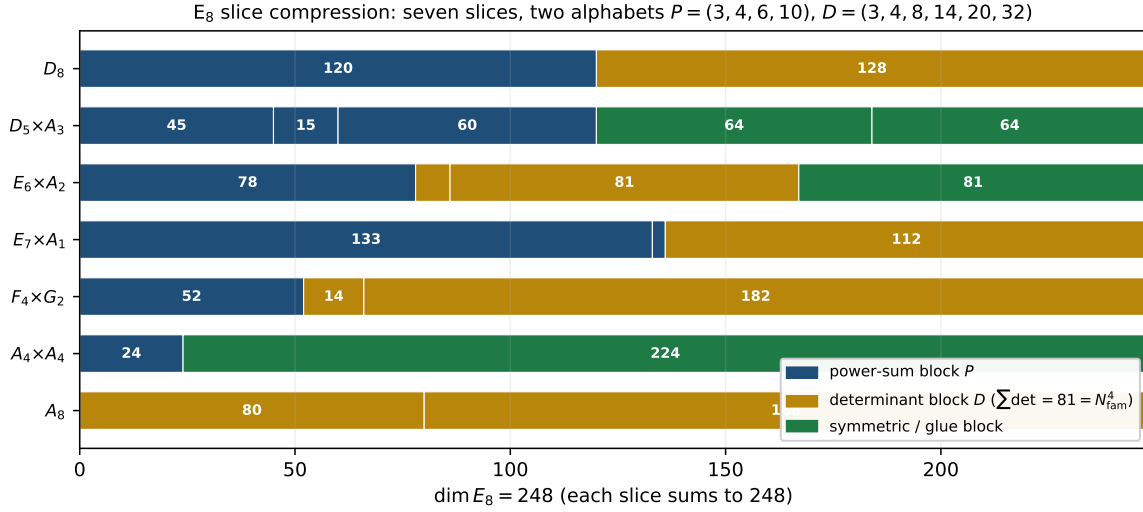


Figure 2: The same seven 248-slices, now coloured by their *alphabet* source rather than by physics module: blue = anchor power-sum blocks ( $P = (3, 4, 6, 10)$ ), gold = Sheet-Diamond determinant blocks ( $D = (3, 4, 8, 14, 20, 32)$ ,  $\sum \det = 81 = N_{\text{fam}}^4$ ), green = symmetric/glue data. Every slice is one projection of the two alphabets ([[verification/v170\\_diamond\\_slice\\_compression.py](#)]).

where  $18 = N_{\text{fam}} |R^+(A_3)|$ . The scalaron exponent 7 (the seam power  $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^7$ ) is thus exactly the  $E_7$  off-block coefficient. *Cross-check:* the centred  $E_8$  exponents give  $\sum_{m \in \text{Exp}(E_8)} |m - 15| = 56 = \dim \mathbf{56}_{E_7}$  (exponents  $\{1, 7, 11, 13, 17, 19, 23, 29\}$ , the primitive residues mod 30).

**Proposition 1** ( $A_4 \times A_4$  action-ladder and  $F_4 \times G_2$  occupancy [O]). 248 = 24 + 24 + 4(5 · 10) with 24 + 24 = 48 =  $\Omega_{\text{adm}}$  and off-diagonal reps (5, 10) =  $(\mathcal{A}_H, \mathcal{A}_\Lambda)$  — the action ladder appears inside  $E_8$  (an audit, *not* an  $SU(5)$  GUT). And  $F_4 \times G_2$ :  $|R(F_4)| = 48 = \Omega_{\text{adm}}$ ,  $|R(G_2)| = 12 = \dim \mathfrak{g}_{\text{SM}}$ , rank  $F_4 = 4 = |\mu_4|$ , rank  $G_2 = 2 = |\mathbb{Z}_2|$ .

**Flavor-matrix and prime-table fingerprints (audit-level, machine-verified).** Five exact linear-algebra readouts of  $R, L$  and the prime atoms, kept here under the no-free-pattern rule.

- **Row norms ladder.**  $\|R_u\|^2, \|R_d\|^2, \|R_e\|^2 = (10, 30, 38) = (\mathcal{A}_\Lambda, h(E_8), 2 \cdot 19)$  with  $10+30+38 = 78 = \dim E_6$  ( $19 \in \text{Exp}(E_8)$ ).
- **Residue inventory.** The entries of  $R$  on  $C_6$  are  $m = (1, 2, 2, 2, 0, 2)$ : the *only* gap is  $4 = |\mu_4| = \sum a$ . Moments  $\sum R = 22$ ,  $\sum R^2 = 78 = \dim E_6$  — so  $\|R\|_F^2 = 78$  is the second moment of the  $C_6$  inventory with the  $\mu_4$  hole.
- **First-column norm**  $\rightarrow E_6 \times A_2$  **off-blocks.**  $L_{\cdot 1} = (7, 7, 8)$ ,  $\|L_{\cdot 1}\|^2 = 162 = 81 + 81$  — the two 81 off-blocks of the  $E_6 \times A_2$  slice are exactly the wound first-generation column.
- **Lepton exponent polynomial.** The  $\varphi_0^{\text{ret}}$ -ladder lepton powers  $(k_e, k_\mu, k_\tau) = (5, 3, 2)$  are the Coxeter primes  $30 = 2 \cdot 3 \cdot 5$  in hierarchy order:  $(t-5)(t-3)(t-2) = t^3 - 10t^2 + 31t - 30$  ( $e_1 = \mathcal{A}_\Lambda$ ,  $e_3 = h(E_8)$ ). The lepton  $c$ -ratios obey  $\frac{(16/7)(7/6)}{4/3} = 2 = |\mathbb{Z}_2|$  and  $\frac{(16/7)(7/6)}{(4/3)^2} = \frac{3}{2}$  (the near-Koide factor).
- **The 2, 3, 5 multiplication table** (the shortest integer-landscape compression):

$$(2, 3, 5) \xrightarrow{\times} (6, 10, 15, 30) \rightarrow (8=3+5, 16, 240=8 \cdot 30, 248=240+8).$$

## 5 Group-theoretic audits (optional)

**Weyl completion index.** Over the main stabiliser,

$$\frac{|W(E_8)|}{|W(D_5)| |W(A_3)|} = \frac{696729600}{1920 \cdot 24} = 15120 = 63 \cdot 240 = (2^6 - 1) |R(E_8)|.$$

A group-theory machine-check, not physics: it quantifies the Weyl symmetry created beyond the  $D_5 \times A_3$  stabiliser. [E] (arithmetic) / [O] (reading)

**$E_8$  theta series as a higher-shell raster.**  $\Theta_{E_8} = E_4 = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$  has shell degeneracies 240, 2160, 6720, 17520, ... The first is  $|R(E_8)| = 240$ ; the second is  $2160 = 9 \cdot 240 = N_{\text{fam}}^2 |R(E_8)|$ . The prime shells read the sheet-generator bilinears of v410:  $\sigma_3(2) = 9 = a^T V \mathbf{1} = N_{\text{fam}}^2$ ,  $\sigma_3(3) = 28 = \mathbf{1}^T V^3 a = \det(I+R)$  and  $\sigma_3(5) = 126$  — the local  $E_7$  root shell  $\{0:126\}$  of lemma (v) below. If later one-loop or CMB shell degeneracies touch these multiplicities, the theta series is the right place to type them — as an audit raster, not a claim. [O] ([verification/v410\_sheet\_generator\_binary.py])

## 6 Further audit lemmas (machine-verified this round)

Five more exact identities tie the flavor matrices and the  $E_8$  exponents to the compiler. All checked with exact integer arithmetic.

**(i) Full-length Frobenius norm is three  $E_6$  blocks.**  $\|L\|_F^2 = 234 = N_{\text{fam}} \|R\|_F^2 = 3 \dim E_6$ , and  $\|L\|_F^2 = \|R\|_F^2 + 12\langle R, W \rangle + 36\|W\|_F^2 = 78 + 48 + 108$ , with  $\langle R, W \rangle = 4 = |\mu_4|$  and  $\|W\|_F^2 = 3 = N_{\text{fam}}$ . The full word-length load is exactly the residue  $E_6$  norm + the  $A_3$ -winding over  $\mu_4$  + the family-winding square. [E]

**(ii)  $A_3$ -winding determinant lemma.** The rank-one winding  $L = R + 6 \mathbf{1} e_1^\top$  and the matrix-determinant lemma with  $e_1^\top R^{-1} \mathbf{1} = \frac{1}{4}$  give  $\det L = \det R (1 + \frac{|R^+(A_3)|}{|\mu_4|}) = 8(1 + \frac{6}{4}) = 20 = 2\mathcal{A}_\Lambda$ .

So  $h(D_5) = 8 \xrightarrow{A_3^+/\mu_4} 20 = |R(A_4)|$ : the winding injects the carrier factor 5 into the cokernel ( $\text{coker } R \simeq \mathbb{Z}_8$ ,  $\text{coker } L \simeq \mathbb{Z}_{20}$ ). [E]

**(iii) The scalaron 7 is the  $E_8$  exponent variance per decuple.** Centring the exponents at  $h/2 = 15$ ,  $\sum_m (m - 15)^2 = 560$  and  $\text{Var} = \frac{560}{8} = 70 = 7 \cdot \mathcal{A}_\Lambda = (\text{scalaron } 7) \cdot (\text{decuple } 10)$ . The halved positive offsets  $\{1, 2, 4, 7\}$  satisfy  $\sum = 14 = 2 \cdot 7$ ,  $\sum^2 = 70$ ,  $\prod = 56 = \dim \mathbf{56}_{E_7}$  — a micro-code carrying the scalaron, the decuple and the  $E_7$  minuscule at once. Also  $\sum_{\text{pairs}} m(30 - m) = 620 = \frac{h \dim E_8}{12} = 2 \cdot 10 \cdot 31$ . [E]

**(iv) Exponents + degrees give the 120+128 split.** The invariant degrees  $d_i = m_i + 1 = \{2, 8, 12, 14, 18, 20, 24, 30\}$  obey  $\sum_i m_i = 120$ ,  $\sum_i d_i = 128$ ,  $248 = 120 + 128$ ,  $\prod_i d_i = |W(E_8)|$ , matching the  $D_8$  branching  $248 = 120 + 128$ ; the degrees are a TFPT dictionary (2, 8, 12, 14, 18, 20, 24, 30 = sheet,  $h(D_5)$ ,  $|R(A_3)|$ ,  $\dim G_2$ ,  $p_4$ ,  $\det L$ ,  $\dim A_4$ ,  $h(E_8)$ ). [E]

**(v) Root shell = local  $E_7 \times A_1$ .** Around any  $E_8$  root the inner-product multiplicities are  $\{+2:1, +1:56, 0:126, -1:56, -2:1\}$ , i.e. the local  $248 = (133, 1) + (1, 3) + (56, 2)$ . A distinguished root (seam/Higgs direction) leaves an orthogonal  $E_7$  and an active coupling shell  $56+56$ . [E]

## 7 What this changes, and what stays open

$E_8$  is one container whose maximal subalgebras each project a different TFPT module:

**Net.** The atlas turns isolated fingerprints (e.g.  $\|R\|_F^2 = 78 = \dim E_6$ , the scalaron 7) into *branching projections* of one container. The discipline rule — “every load-bearing number must appear in  $\geq 1$  projection” — makes the structure falsifiable: a number in no projection is suspect. This is a *program*, not a proof of new physics: all readings are [O]. The two genuinely load-bearing, derived results remain the  $\mu_4$  glue ( $E_8 = (D_5 \oplus A_3) + \mu_4$ , [E] [verification/v1\_e8\_glue.py]) and the spectral selector of the flavor matrix ( $\det R = h(D_5)$ ,  $\text{PrinMin}_2 = (2, 3, 5)$ , [E] [verification/v4\_flavor\_matrix.py]). The remaining flavor-geometry step (historically “H2”) is sharpened in the parabolic note to: *the  $D_4$ -equivariant stable parabolic structure on  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  realises the unique branch with  $\det R = 8$  and  $\text{PrinMin}_2(R) = (2, 3, 5)$  — no longer “find the matrix”, but “prove this one geometric realisation”.* (Status: since v118–v122 the dictionary *values* are exact theorems — hexagon = sign-twisted family spectrum, addresses pinned, margins forced; what remains is exactly the realisation statement above, the selector establishment R4’.)

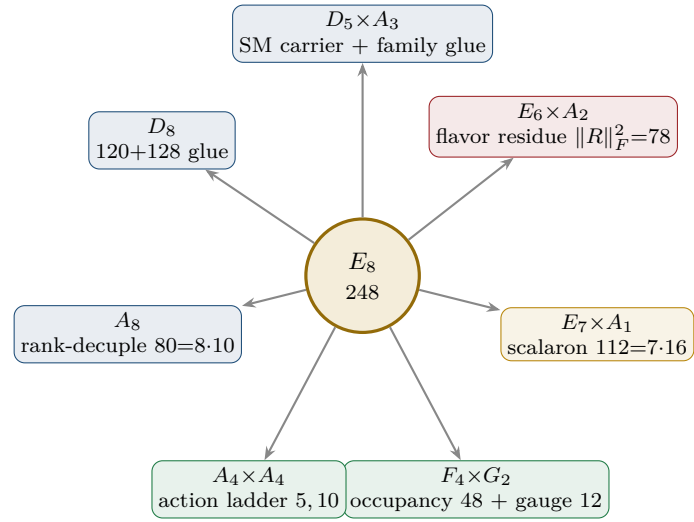


Figure 3:  $E_8$  as an audit container: each maximal subalgebra projects one TFPT module. The discipline rule — every load-bearing number appears in  $\geq 1$  projection — makes the structure falsifiable.

## Part II

# The $E_8$ cascade bridge

**The connection in one sentence.** The old  $E_8$  cascade (Paper V1.06: the nilpotent-orbit chain  $D_n = 60 - 2n$ ,  $n = 0 \dots 26$ ) and the new compiler/atlas are *the same even-integer  $E_8$  spine*, read three ways: the cascade *orders* the numbers  $\{8, 10, \dots, 60\}$  as a scale ladder, the atlas *reads* them as  $E_8$  subalgebra branchings, and the compiler *generates* them from the  $(1, 1, 2)$  anchor. The cascade *starts* at  $D_{\text{start}} = 60 = p_2 p_3$  and *ends* at  $8 = h(D_5) = \det R$  — the flavor selector.

## 8 The old cascade, recalled

Paper V1.06 builds a deterministic scale ladder from the nilpotent orbits of  $E_8$ . With centralizer dimension  $D = 248 - \dim \mathcal{O}$  and the Hasse chain restricted to  $\Delta D = 2$ , a unique maximal chain runs from  $A_4 + A_1$  ( $D_0 = 60$ ) down to the regular orbit ( $D_{26} = 8$ ):

$$D_n = 60 - 2n, \quad n = 0, \dots, 26; \quad \varphi_{n+1} = \varphi_n e^{-\gamma(n)}, \quad \gamma(n) = \lambda [\ln D_n - \ln D_{n+1}]$$

The cascade arithmetic (endpoints, exponent rungs, IR-tail product, variance) is machine-checked. `[verification/v5_e8_cascade.py]` with the single structural normalisation  $\lambda = \gamma(0)/(\ln 248 - \ln 60) = 0.58770$ , fixed by a curvature-extremum on the chain (no fit). The  $E$ -windows of the two-loop flow anchor it:  $\alpha_3 = 1/(8\pi) = c_3$  (the  $E_8$  window) and  $\alpha_3 = 1/(6\pi)$  (the  $E_6$  window, near  $\varphi_0^{\text{ret}}$ ).

## 9 The spine: every rung is a current number

Reading  $D = 60 - 2n$  against today's compiler atoms, the secondary-branching atlas and the  $(1, 1, 2)$  anchor power sums  $p_n = 2 + 2^n$ , **25 of the 27 rungs are load-bearing numbers** (machine-checked):

| $n$ | $D$ | current meaning                                                                | $n$ | $D$ | current meaning                                          |
|-----|-----|--------------------------------------------------------------------------------|-----|-----|----------------------------------------------------------|
| 0   | 60  | $D_{\text{start}} = p_2 p_3 = \mathcal{A}_\Lambda  R^+(A_3) $ ( <b>start</b> ) | 13  | 34  | $p_5$ anchor power = $2 \cdot 17$                        |
| 1   | 58  | $2 \cdot 29$ ( $E_8$ exponent)                                                 | 14  | 32  | $2^5 = 2 \dim S^+$                                       |
| 2   | 56  | $\dim \mathbf{56}_{E_7} = a^\top L \mathbf{1} = \sum  m-15 $                   | 15  | 30  | $h(E_8) = 2 \cdot 3 \cdot 5$ ( <b>Coxeter rung</b> )     |
| 3   | 54  | $2 \cdot 27$ , $27 = \mathbf{1}^\top R a = 3^3$                                | 16  | 28  | $4 \cdot 7$ (half of 56)                                 |
| 4   | 52  | $\dim F_4 =  R(D_5)  +  R(A_3) $                                               | 17  | 26  | $2 \cdot 13$ (old hadronic block)                        |
| 5   | 50  | $A_4 \times A_4$ block = $5 \cdot 10$                                          | 18  | 24  | $\dim A_4$ ; $24+24=48$                                  |
| 6   | 48  | $\Omega_{\text{adm}} =  R(F_4)  = 2p_1 p_2$                                    | 19  | 22  | $\sum R = \mathbf{1}^\top R \mathbf{1}$ (residue sum)    |
| 7   | 46  | $2 \cdot 23$ ( $E_8$ exponent)                                                 | 20  | 20  | $\det L =  R(A_4)  = 2\mathcal{A}_\Lambda$               |
| 8   | 44  | $4 \cdot 11$ ( $E_8$ exponent)                                                 | 21  | 18  | $p_4 = N_{\text{fam}}  R^+(A_3) $                        |
| 9   | 42  | $6 \cdot 7 =  R^+(A_3)  \cdot 7$                                               | 22  | 16  | $\dim S^+$ (one generation)                              |
| 10  | 40  | $ R(D_5)  = \sum L = \mathbf{1}^\top L \mathbf{1}$                             | 23  | 14  | $\dim G_2 = \sum L_d = 8+6$                              |
| 11  | 38  | $2 \cdot 19$ ( $E_8$ exponent)                                                 | 24  | 12  | $ R(A_3)  = \dim \mathfrak{g}_{\text{SM}}$               |
| 12  | 36  | $6^2$ (old EW block)                                                           | 25  | 10  | $\mathcal{A}_\Lambda = \binom{5}{2}$ (old cosmo block)   |
|     |     |                                                                                | 26  | 8   | $h(D_5) = \det R = 2p_1$ ( <b>end: flavor selector</b> ) |

Three structural facts jump out (all **[E]** arithmetic):

- **Endpoints are the compiler boundary.** The cascade *starts* at  $60 = p_2 p_3 = D_{\text{start}}$  and *ends* at  $8 = h(D_5) = \det R$  — the flavor spectral selector is literally the deepest orbit of  $E_8$ .
- **Coxeter rung is  $h(E_8)$ .** At  $n = 15$ ,  $D = 30 = h(E_8)$  — this is the *Coxeter rung*, not the arithmetic centre; the median node is  $n = 13$ ,  $D = 34 = p_5(a)$  (below). It matches the exponent-centering audit  $\sum_m |m - 15| = 56 = \dim \mathbf{56}_{E_7}$  (which is itself rung  $n = 2$ ).
- **Doubled  $E_8$  exponents are rungs.**  $\{58, 46, 38, 34, 26, 22, 14\} = 2 \times \{29, 23, 19, 17, 13, 11, 7\}$  are exactly the cascade rungs at  $n = 1, 7, 11, 13, 17, 19, 23$ ; three of them coincide with anchor

numbers ( $34 = p_5$ ,  $22 = \sum R$ ,  $14 = \dim G_2$ ). Only  $n = 8$  ( $D = 44 = 4 \cdot 11$ ) is not independently labelled.

### Cascade arithmetic — endpoints, centre and length (verified)

The 27 rungs encode  $E_8$  in their endpoints, and their centre is the anchor:

$$\begin{aligned} \frac{D_{\text{start}} D_{\text{end}}}{2} &= \frac{60 \cdot 8}{2} = 240 = |R(E_8)|, & 240 + D_{\text{end}} &= 248 = \dim E_8, & D_{\text{start}} &= 2h(E_8) = 60; \\ \text{arithmetic median} &= \frac{60+8}{2} = 34 = p_5(a) = h(E_8) + |\mu_4|, & \sum_{n=0}^{26} (D_n - 30) &= 27 \cdot 4 = 27|\mu_4|; \\ \text{width} = D_{\text{start}} - D_{\text{end}} &= 52 = \dim F_4, & 60 \rightarrow 30 &= h(E_8), & 30 \rightarrow 8 &= \sum R = 22; \\ \#\text{nodes} = 27 &= 3^3 = \mathbf{1}^\top Ra \text{ (} E_6 \text{ cubic block)}, & \#\text{steps} &= 26 = \dim \mathbf{26}_{F_4}. \end{aligned}$$

So the cascade is an  $E_8$  Coxeter ladder centred on  $p_5(a)$  with a per-node  $|\mu_4|$  glue shift; its length (27 nodes, 26 steps) carries the  $E_6$  cube and the  $F_4$  fundamental. [E]

### The IR tail $n = 24 \dots 30$ : the compiler atoms

Continuing  $D = 60 - 2n$  past the last orbit ( $D=8$ ,  $n=26$ ) is no longer a nilpotent-orbit chain but the *algebraic compiler tail* of the Coxeter clock:

$$D_{24 \dots 30} = \underbrace{12}_{|R(A_3)|=\dim \mathfrak{g}_{\text{SM}}}, \underbrace{10}_{\mathcal{A}_\Lambda}, \underbrace{8}_{h(D_5)}, \underbrace{6}_{|R^+(A_3)|}, \underbrace{4}_{|\mu_4|}, \underbrace{2}_{\text{sheet}}, \underbrace{0}_{\text{closure}}.$$

This is exactly what the old “ $n = 0 \dots 30$  clock” was scanning:  $n=0 \dots 26$  the  $E_8$  orbit spine,  $n=27 \dots 30$  the compiler atoms. (Mark the tail as the *compiler tail*, not orbits.) [E]



**Three further cascade lemmas (all machine-verified)**

The clock carries more  $E_8$  data than the endpoints:

- **Primitive rungs sum to the root count.** Evaluating  $D_n = 60 - 2n$  at the *primitive* indices  $n \in (\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\} = \text{Exp}(E_8)$  gives  $\{58, 46, 38, 34, 26, 22, 14, 2\}$  with

$$\sum_{n \in \text{Exp}(E_8)} (60 - 2n) = 240 = |R(E_8)|,$$

and since  $30 - n$  is primitive when  $n$  is, the clock indexes its own  $E_8$  exponents.

- **The IR tail product is the main Weyl stabiliser.** The nonzero tail  $\{12, 10, 8, 6, 4, 2\}$  has

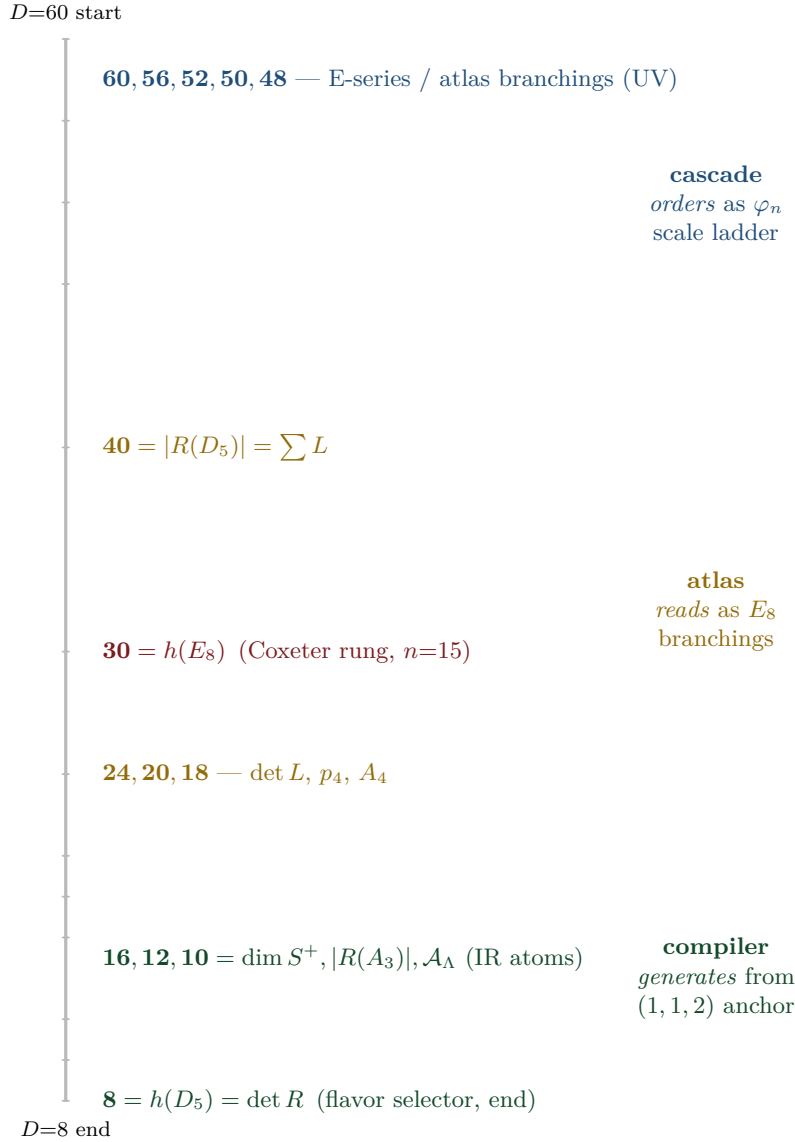
$$\prod = 46080 = |W(D_5)| |W(A_3)| = 1920 \cdot 24, \quad \sum = 42 = |R^+(A_3)| \cdot 7.$$

- **Cascade variance =  $E_6$  flavour norm times the scalaron-gauge block.** The 27 rungs  $60, \dots, 8$  have mean  $34 = p_5(a)$  and

$$\sum_{n=0}^{26} (D_n - 34)^2 = 6552 = 78 \cdot 84 = \dim E_6 \cdot (7 \dim \mathfrak{g}_{\text{SM}}),$$

tying the cascade simultaneously to the  $E_6 \times A_2$  flavour shadow and the  $A_8$  scalaron-gauge shadow.  
**[E]** (audit level)

## 10 Three views of one spine



**The unification.** The same numbers  $\{8, 10, \dots, 60\}$  are: (a) the *scale ladder*  $\varphi_n$  of the old cascade (orbit dimensions, ordered by  $e^{-\gamma}$ ); (b) the *branching blocks* of the secondary-atlas ( $\dim F_4=52$ ,  $\dim \mathbf{56}_{E_7}$ ,  $5 \cdot 10, \dots$ ); (c) the *anchor power sums and matrix invariants* of the compiler ( $p_n = 2 + 2^n$ ,  $\det R=8$ ,  $\sum L=40$ ). The cascade is the *UV*→*IR* ordering; the compiler is its *IR endpoint*, where the small numbers 8, 10, 12, 16 are the flavor/family/carrier data.

**The spine tetrahedron: the microkernel as one finite object [E]**  
( [verification/v91\_spine\_tetrahedron.py] )

The spine  $T = \{2, 3, 4, 5\} = \{\mathbb{Z}_2, N_{\text{fam}}, |\mu_4|, g_{\text{car}}\}$  equals  $\{e_3(a), p_0(a), e_1(a), e_2(a)\}$  for the anchor  $a = (1, 1, 2)$  — the tetrahedron on these four nodes is the anchor plus its zeroth moment, not a second structure. Its combinatorics *are* the central integer grammar: edge products  $\{6, 8, 10, 12, 15, 20\}$  ( $|R^+(A_3)|$ ;  $\text{rank } E_8=h(D_5)=\det R$ ;  $\mathcal{A}_\Lambda=|E(K_5)|$ ;  $|R(A_3)|=\dim \mathfrak{g}_{\text{SM}}$ ;  $\dim A_3$ ;  $\det L$ ), face products  $\{24, 30, 40, 60\}$  ( $|W(A_3)|$ ;  $h(E_8)$ ;  $|R(D_5)|=\sum L$ ;  $D_{\text{start}}$ ), volume  $2 \cdot 3 \cdot 4 \cdot 5 = 120 = |R^+(E_8)| = 5!$ ; and the graph reading  $|R(E_8)| = 240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$  with  $|E(K_4)| = p_2 = 6$ ,  $|E(K_5)| = p_3 = 10$  — including the honest negative control that the identification *breaks* at  $K_6$  ( $|E(K_6)| = 15 \neq p_4 = 18$ ), so it is specific, not generic. Two disciplined caveats: the “dual cuts” (node  $\times$  opposite face = 120, etc.) are tautologies of any complementary 4-set partition and are typed as presentation only; and the tetrahedron grammar is a *sub*-grammar — the load-bearing 7, 16, 41, 48, 240, 248 are *not* subset products and still require the

anchor power sums / Pascal / glue layers.

## 11 Direct $\varphi_0^{\text{ret}}$ relations: old vs current

The old cascade’s “fundamental relations near  $n = 0$ ” connect directly to the current note — one is *identical*, two are early forms of current refinements:

| Quantity        | old (V1.06)                                  | current                                                                          | relation                                                |
|-----------------|----------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|
| $\Omega_b$      | $\varphi_0^{\text{ret}}(1 - 2c_3) = 0.04894$ | $(1 - \frac{1}{4\pi})\varphi_0^{\text{ret}} = 0.04894$                           | <b>identical</b> ( $2c_3 = \frac{1}{4\pi}$ ) [E]        |
| $V_{us}/V_{ud}$ | $\sqrt{\varphi_0^{\text{ret}}} = 0.2306$     | $\lambda_C = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})} = 0.2244$ | refined (the $(1 - \varphi_0^{\text{ret}})$ factor) [E] |
| $r$ (tensor)    | $(\varphi_0^{\text{ret}})^2 = 0.00283$       | $12/N_\star^2 = 0.00397$                                                         | two branches, same order [C]                            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

So  $\Omega_b = \varphi_0^{\text{ret}}(1 - 2c_3)$  was already the exact current formula in V1.06; the Cabibbo ratio gained its  $(1 - \varphi_0^{\text{ret}})$  factor; and the old direct  $r = (\varphi_0^{\text{ret}})^2$  and the current Starobinsky  $r = 12/N_\star^2$  agree in order ( $\sim 0.003$ ) but are distinct predictions — a clean, testable discriminant, stated honestly.

## 12 Sector inventory: have all the old-cascade sectors been computed?

The old cascade also carried *calibrated absolute scales*: per block, one unit  $\zeta_B$  was fixed on a single reference and every other scale in the block followed fit-free from the ratio law  $\varphi_m/\varphi_n = (D_m/D_n)^\lambda$  (Paper V1.06, App. C). These block assignments are the place where the proton mass, the EW masses and the CMB temperature lived. The honest question is: *has the new compiler re-derived each of these, or only some?* The full inventory, with the current status spelled out:

| Old block (rung)     | Old observable & calibration                                                                                                             | Current-model status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mark |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| near $n=0$           | $\Omega_b = \varphi_0^{\text{ret}}(1 - 2c_3),$<br>$V_{us}/V_{ud} = \sqrt{\varphi_0^{\text{ret}}},$<br>$r = (\varphi_0^{\text{ret}})^2$   | <b>covered</b> — dimensionless, derived from $\{c_3, \varphi_0^{\text{ret}}\}$ ; see the $\varphi_0^{\text{ret}}$ -table above. $\Omega_b$ is <i>identical</i> .                                                                                                                                                                                                                                                                                                                                                   | [E]  |
| EW, $n=12$           | $v_H = \zeta_{\text{EW}} M_{\text{P}1} \varphi_{12},$<br>$M_W = \frac{1}{2} g_2 v_H, M_Z$                                                | <b>scale covered, absolute scheme-level</b> — the EW scale enters the current note via the action-grammar exponent and the two-loop flow; $M_W, M_Z$ are dimensionful and remain RG/scheme outputs (one $\zeta_{\text{EW}}$ ), <i>not</i> new compiler powers.                                                                                                                                                                                                                                                     | [C]  |
| hadronic, $n=15, 17$ | $m_p = \zeta_p M_{\text{P}1} \varphi_{15},$<br>$m_b = \zeta_b M_{\text{P}1} \varphi_{15},$<br>$m_u = \zeta_u M_{\text{P}1} \varphi_{17}$ | <b>ratios closed, absolute scale an anchor.</b> The quark mass <i>ratios</i> sit on the $\varphi_0^{\text{ret}}$ -word-length ladder and are closed (integer Plücker readouts, $\sqrt{49/\sqrt{71}}$ ); the <i>absolute</i> values need the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). The <b>proton mass</b> is QCD-confinement, $m_p \sim \Lambda_{\text{QCD}}$ by dimensional transmutation — a <i>cross-sector</i> ratio, listed as a genuine frontier item ( $m_p/m_e, \text{tfpt\_4}$ ). | [O]  |
| cosmo, $n=25$        | $T_{\gamma 0} = \zeta_\gamma M_{\text{P}1} \varphi_{25},$<br>$T_\nu = (\frac{4}{11})^{1/3} T_{\gamma 0}, \rho_\Lambda$                   | <b>partly.</b> $\rho_\Lambda/\Lambda$ is <i>typed</i> by the action grammar and the seam ( $\Lambda$ -typing, $\text{tfpt\_2}$ ); $T_\nu/T_{\gamma 0}$ is standard QFT (not cascade). The <b>absolute CMB temperature</b> $T_{\gamma 0}$ is a downstream readout of the reheating pipeline (the old cosmology/CMB layers; today the scalaron-reheating chain, $\sqrt{v86}$ ), conditional — not re-derived as a compiler number.                                                                                   | [C]  |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

**Direct answer: which sectors are closed, which are not**

**Closed (derived from  $\{c_3, g_{\text{car}}, \varphi_0^{\text{ret}}\}$ ):** the dimensionless near- $n=0$  anchors ( $\Omega_b$  exactly, Cabibbo, tensor branch) and all the *dimension/rung* structure of the spine. **Scale-level only (one  $\zeta_B$  each, scheme/RG):** the EW masses  $M_W, M_Z$  and the cosmological absolute scales  $T_{\gamma 0}$  — exactly as in the current SM/cosmology notes; these were *never* pure predictions, not even in V1.06 (each needed a block unit). **Genuinely open:** the **proton mass**  $m_p$  (and  $m_p/m_e$ ) is QCD-confinement across two sectors and is *not* claimed as a compiler power — it stays a frontier item ([O], tfpt\_4); the quark mass *ratios* are closed (integer Plücker,  $\sqrt{49}/\sqrt{71}$ ) and only the *absolute* quark scale  $m_b, m_u$  needs the per-sector  $\Lambda/U_f^*$  normalisation, an anchor ([O]). So: yes, the *spine* is fully audited and the *dimensionless* sectors are derived; the *dimensionful* hadronic/cosmo absolute scales are honestly still scheme-level, with the proton mass the one explicitly open cross-sector number — nothing is silently claimed.

## 13 What is new, what is old, what stays open

**New structures found via the bridge**

1. The flavor selector  $\det R = h(D_5) = 8$  is the *terminal orbit* of the  $E_8$  cascade; the compiler is the cascade's IR endpoint. [E]
2. The atlas branching blocks (52, 56, 60) are the cascade's *UV head* ( $n = 0, 2, 4$ ); the secondary atlas and the orbit cascade are the same spine read top-down vs. as subalgebras. [E]
3. The anchor power sums sit on doubled-exponent rungs:  $p_5 = 34$  ( $n=13$ ),  $\sum R = 22$  ( $n=19$ ),  $\dim G_2 = 14$  ( $n=23$ ). [E]
4. The cascade Coxeter rung  $D = 30 = h(E_8)$  ( $n=15$ ) ties to the exponent-centering audit  $\sum |m-15| = 56$ . [E]

**Honest separation**

The *dimension/rung coincidences* are exact arithmetic [E] — they show the old and new constructions live on one  $E_8$  spine. The old cascade's *scale assignments*  $\varphi_n$  (EW at  $n=12$ , hadronic at  $n=15, 17$ , cosmo at  $n=25$ ) are a *different* mechanism from the current  $\varphi_0^{\text{ret}}$ -power mass ladder: the cascade orders *orders of magnitude* (which sector sits where) via  $e^{-\gamma}$ , while the  $\varphi_0^{\text{ret}}$ -ladder gives *values within* a sector. Both are real, complementary orderings; neither is forced onto the other. The per-sector  $\zeta$ -calibrated absolute scales (proton mass, EW masses, CMB temperature) remain scheme-level — audited one by one in §12, with the proton mass the single explicitly open cross-sector number.

**Bottom line.** The old  $E_8$  cascade was an early, correct intuition: the physically relevant numbers are exactly the even-integer  $E_8$  orbit spine  $\{8, \dots, 60\}$ . The current work *explains why* — they are the  $(D_5 \times A_3) + \mu_4$  compiler atoms and their  $E_8$  branching blocks — and identifies the spine's two ends as the compiler boundary ( $D_{\text{start}} = 60$ ,  $\det R = 8$ ). Old and new are the same structure at different resolution.

## Part III

# The self-consistency loop

**The question, and the honest answer.** The theory rests on two inputs,  $c_3 = \frac{1}{8\pi}$  and  $g_{\text{car}} = 5$ . The old theory had a Möbius seam and the seed was the *start* of the  $E_8$  cascade. Run it *backwards*: can the cascade/ $E_8$  *return*  $c_3$  and  $g_{\text{car}}$ , closing a self-consistent loop? **Answer:** the *discrete* content is a closed, overdetermined fixed point —  $g_{\text{car}} = 5$  and the “8” in  $c_3$  are forced by the  $E_8$  closure. The *only* irreducible primitive left is the geometric  $\pi$  (the Möbius/boundary Gauss–Bonnet). So “two axioms” collapses to “*one* continuous primitive  $\pi$  + *one* discrete self-consistency fixed point.”

## 14 The loop for $g_{\text{car}}$ — an overdetermined fixed point

Two *independent* structural conditions, both pure Lie-theory, force  $g_{\text{car}} = 5$  once  $E_8$  is the closure object and  $A_3$  is the four-puncture family geometry:

### $g_{\text{car}} = 5$ is forced twice over

(A) rank fill.

The glue  $D_{g_{\text{car}}} \oplus A_3 \hookrightarrow E_8$  must fill the rank:  $\text{rank } D_{g_{\text{car}}} + \text{rank } A_3 = \text{rank } E_8$ , i.e.  $g_{\text{car}} + 3 = 8 \Rightarrow g_{\text{car}} = 5$ .

(B) Coxeter match.

The carrier Coxeter number must equal the  $E_8$  rank,  $h(D_{g_{\text{car}}}) = 2g_{\text{car}} - 2 = 8 \Rightarrow g_{\text{car}} = 5$ .

Both give  $g_{\text{car}} = 5$ , and the loop *closes*:  $g_{\text{car}} = 5 \Rightarrow D_5 \oplus A_3 \xrightarrow{\mu_4} E_8 \Rightarrow h(E_8) = 30 = 2 \cdot 3 \cdot 5 \Rightarrow \text{largest prime} = 5 = g_{\text{car}}$ . [E]

### Reverse glue selection — $D_5$ and $A_3$ from unimodular closure

The closure even *selects the two factors*, not just the rank. The discriminant of  $A_{\mu-1}$  is  $\mathbb{Z}_\mu$  and of  $D_g$  (odd  $g$ ) is  $\mathbb{Z}_4$ ; a common cyclic glue forces  $\mu = 4$ , hence  $A_{\mu-1} = A_3$ . The even-glue norm  $q(D_g) + q(A_3) = \frac{g}{4} + \frac{3}{4} = 2$  then forces  $g = 5$ , hence  $D_5$ . A second, independent confirmation of  $\mu = 4$  is the *mirror identity*  $\dim D_{\mu+1} + \dim A_{\mu-1} = \dim V_{D_{\mu+1}} \cdot |R^+(A_{\mu-1})|$ , i.e.  $3\mu(\mu+1) = \mu(\mu+1)(\mu-1) \Rightarrow \mu-1 = 3 \Rightarrow \mu = 4$  (the  $45 + 15 = 10 \cdot 6$  identity). So

$$\text{common } \mathbb{Z}_4 \text{ glue} + \text{root norm } 2 \implies D_5 + A_3$$

—  $g_{\text{car}} = 5$  is recovered from the closure, not just posited. [E]

### Familyful $E_8$ -closure fixed-point theorem

Among rank-filling pairs  $D_g \oplus A_m$  ( $g+m=8$ ) admitting an integer glue index  $\sqrt{\det D_g \det A_m} = \sqrt{4(m+1)}$ , exactly two close to an even unimodular lattice:  $D_8 \oplus A_0$  (no family) and  $D_5 \oplus A_3$  (familyful). The unique familyful solution is  $D_5 \oplus A_3$ , with lattice  $\cong E_8$  and  $g_{\text{car}} = 5$ .

| $D_g + A_m$                   | $g+m$ | glue idx <sup>2</sup> =4(m+1) | integer?                      | status                                    |
|-------------------------------|-------|-------------------------------|-------------------------------|-------------------------------------------|
| $D_8 + A_0$                   | 8     | 4                             | 2                             | valid, <i>no family</i>                   |
| $D_7 + A_1$                   | 8     | 8                             | —                             | no integer glue                           |
| $D_6 + A_2$                   | 8     | 12                            | —                             | no integer glue                           |
| <b><math>D_5 + A_3</math></b> | 8     | 16                            | <b>4= <math>\mu_4</math> </b> | <b>valid, familyful, <math>E_8</math></b> |
| $D_4 + A_4$                   | 8     | 20                            | —                             | no integer glue                           |

Equivalently, with  $q(A_{\mu-1}) = \frac{\mu-1}{\mu}$  and  $g = 9 - \mu$  the norm closure  $q(D_g) + q(A_{\mu-1}) = 2$  becomes  $\mu^2 - 5\mu + 4 = 0$ , whose nontrivial root is  $\mu = 4$  ( $A_3, g = 5$ ). So  $g_{\text{car}} = 5$  is forced *three* ways: rank-fill, Coxeter-match, and integer-glue/norm closure. [E] [verification/v6\_bootstrap.py]

**Bootstrap classification:  $D_5 \oplus A_3$  is the *unique* familyful  $E_8$  glue [E]**

$E_8$  is the unique even unimodular rank-8 lattice. Its two-factor root-lattice gluings  $L_1 \oplus L_2$  (rank = 8) that admit a *cyclic* glue are exactly those whose discriminant groups are isomorphic and cyclic:

$$D_5 \oplus A_3 (\mathbb{Z}_4), \quad E_6 \oplus A_2 (\mathbb{Z}_3), \quad E_7 \oplus A_1 (\mathbb{Z}_2), \quad A_4 \oplus A_4 (\mathbb{Z}_5).$$

Imposing the two TFPT data — one factor with a 16-dimensional half-spinor ( $\dim S_{D_g}^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ ) and one factor giving exactly  $N_{\text{fam}} = 3$  families (rank  $A_m = m = 3$ ) — selects

$$D_5 \oplus A_3 \text{ (uniquely), glue group } \mathbb{Z}_4 = |\mu_4|.$$

$E_6 \oplus A_2$  gives only 2 families,  $E_7 \oplus A_1$  only 1;  $D_8$  has non-cyclic glue  $(\mathbb{Z}_2)^2$  and  $A_8$  is a single rank-8 factor. So the carrier+family decomposition is forced. [verification/v15\_bootstrap\_classification.py]

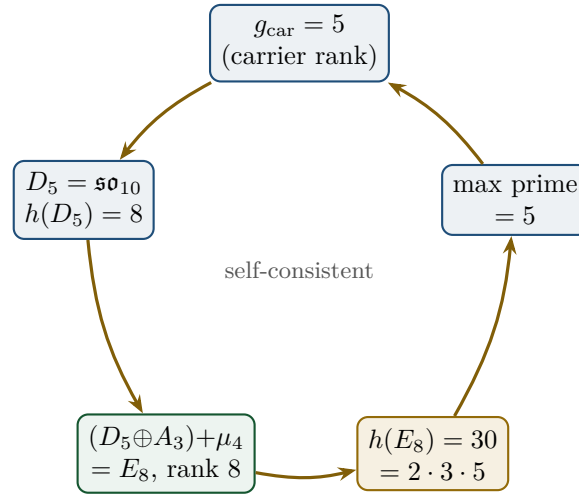


Figure 4: The closed loop:  $g_{\text{car}} = 5$  builds  $E_8$ , whose Coxeter number  $30 = 2 \cdot 3 \cdot 5$  returns the carrier prime 5. The value  $g_{\text{car}} = 5$  is the unique fixed point (overdetermined by rank-fill *and* Coxeter-match).

The carrier split  $3 + 2$  and the one-generation count  $\dim S^+ = 2^{g_{\text{car}}-1} = 16$  are then automatic, as is the glue norm  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ . The family rank is rank  $A_3 = 3 = N_{\text{fam}}$  and the glue index is  $h(A_3) = 4 = |\mu_4|$  — the Coxeter numbers of the three players are  $(h(A_3), h(D_5), h(E_8)) = (4, 8, 30) = (|\mu_4|, \det R, \text{compiler})$ .

## 15 The loop for $c_3$ — the “8” is the $E_8$ rank

The boundary normalisation is  $c_3 = 1/(8\pi)$ . Its integer is exactly the carrier Coxeter number, which the loop above equals to the  $E_8$  rank:

$$c_3 = \frac{1}{h(D_5) \pi} = \frac{1}{(2g_{\text{car}} - 2) \pi} = \frac{1}{8\pi}, \quad 8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$$

where  $\varphi(30) = 8$  is Euler’s totient (the number of primitive residues mod  $30 =$  the  $E_8$  exponents) — so the “8” has *five* concordant readings. The tree seed  $\varphi_{\text{tree}} = 1/(6\pi)$  has  $6 = |R^+(A_3)| = h(E_8)/g_{\text{car}}$ , the three Gauss–Bonnet boundary cycles ( $6\pi = 3 \times 2\pi$ ). So the *discrete* parts of both  $c_3$  and  $\varphi_{\text{tree}}$  are  $E_8/\text{carrier}$  invariants; only the factor  $\pi$  is genuinely continuous.

**Honest caveat on  $8\pi$** 

$8\pi$  is *also* the standard Einstein normalisation ( $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ ). The loop *identifies* this gravitational 8 with rank  $E_8 = h(D_5)$  — a non-coincidence *if*  $E_8$  is the closure, but one should be clear that  $8\pi$  has an independent GR reading. The loop promotes the two 8's to the same number; it does not derive  $\pi$ .

**16 The Möbius origin of the irreducible  $\pi$** 

What the loop does *not* remove is  $\pi$ . In the old theory this is exactly the Möbius datum: the orientable double cover of the Möbius fibre is a cylinder with two boundaries plus one  $\mathbb{Z}_2$  seam  $\Gamma$ ; Gauss–Bonnet gives  $\oint k ds = 2\pi + 2\pi + 2\pi = 6\pi$  (hence  $\varphi_{\text{tree}} = 1/(6\pi)$ ), and the  $\mathbb{Z}_2$  identification is the sheet “2” of  $30 = 2 \cdot 3 \cdot 5$ . So the Möbius seam supplies precisely the two things the discrete loop cannot: the continuous  $\pi$  and the sheet parity  $\mathbb{Z}_2$ .

**Why “Möbius” is not incidental: it is the boundary Lorentz structure.** The Möbius group of the Riemann sphere is *exactly* the proper orthochronous Lorentz group:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ , with the sphere realised as the celestial sphere of null directions via stereographic projection (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, CUP 1984, §1.2–1.4). This is the spacetime meaning of the self-consistency loop: the seam’s boundary-2-sphere carries the Lorentz symmetry, the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation of that sphere (the order-4 Möbius map, [verification/v180\_clock\_is\_mobius.py]), and the  $\text{RP} \rightarrow \text{OS}$ -reconstructed causal cone (the shared  $v_\gamma = v_g = c$  of `tfpt_horizon_readouts`) is the same Lorentz structure read off the seam. So “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$  — not a coincidence of names. [O] ( $\pi$  stays primitive; the identification is a structural reading). The *celestial* reading of this sphere has since been pinned at the exact-arithmetic level and has grown into a programme of its own (the research contract `CELEST.SEAM.01`); it is presented as a dedicated section below — the celestial and twistor continuum route (Section 17).

**Net: two axioms  $\rightarrow$  one primitive + one fixed point.**

$$\{c_3, g_{\text{car}}\} \longrightarrow \underbrace{\pi \text{ (Möbius/Gauss–Bonnet)}}_{\text{one continuous primitive}} + \underbrace{E_8 \text{ closure}}_{\text{one discrete fixed point}}$$

The discrete data  $\{2, 3, 4, 5, 6, 8, 30\}$  are *all* self-consistent outputs of the  $E_8$  closure ( $g_{\text{car}} = 5$ ,  $h(D_5) = 8$ ,  $h(A_3) = 4 = |\mu_4|$ ,  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ ,  $|R^+(A_3)| = 6$ ). The only irreducible input on the *dimensionless* axis is  $\pi$ . [E] (discrete loop) / [O] ( $\pi$  is primitive)

On the *dimensionful* axis there is exactly *one* more irreducible: a single mass scale. But even that is no longer a *free* constant — fixing the physical  $\Lambda$  branch pins  $\bar{M}_{\text{Pl}}$ , so  $G_N$  is a  $\Lambda$ -metrology *output* (`origin_theory`,  $\Lambda$ -typing); the one genuinely irreducible scale is the induced-gravity (Sakharov) anchor. So the complete honest reduction is “ $\pi$  + the  $E_8$ -closure fixed point + *one* dimensionful scale”.

*One sharpening of the “8”.* Both  $\pi$ -coefficients are  $2\pi$  times a compiler integer:  $\varphi_{\text{tree}} = 1/(6\pi)$  with  $6 = 2N_{\text{fam}}$  (three Gauss–Bonnet boundary circles) and  $c_3 = 1/(8\pi)$  with  $8 = 2|\mu_4|$  (seam winding), so  $\varphi_{\text{tree}} = (|\mu_4|/N_{\text{fam}})c_3 = \frac{4}{3}c_3$ . The *same* integer 8 is the Hawking/Einstein coefficient ( $G_{\mu\nu} = 8\pi T_{\mu\nu}$ , Appendix H): the seam normaliser, the  $E_8$  rank and the universal horizon factor are one number, overdetermined ( $8 = 2|\mu_4| = \text{rank } E_8 = h(D_5) = \varphi(30) = \det R$ ). [E] [verification/v54\_seam\_horizon\_keystones.py]

**A cyclic element *inside* the hull: the order-30 Coxeter rotation [E]**  
([verification/v55\_coxeter\_cycle.py])

The compiler hull carries a *literal* cyclic symmetry. Built from the  $E_8$  Cartan matrix, the Coxeter element  $c = s_1 \cdots s_8$  has **order exactly**  $h(E_8) = 30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}} = 2 \cdot 3 \cdot 5$  (verified by direct matrix powers), and its eigenvalues are the primitive 30th roots  $e^{2\pi i m/30}$  with  $m$  the  $E_8$  *exponents*  $\{1, 7, 11, 13, 17, 19, 23, 29\}$  — which are precisely the  $\varphi(30) = 8$  *totatives* of 30. Hence

$$\text{rank } E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}$$



So the 8 in  $c_3 = \frac{1}{8\pi}$  is the count of live phases of a primitive order-30 cyclic rotation of the hull, whose period  $30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}}$  is the product of the three core integers. The cyclic structure is *present in the mathematics*, not interposed.

**The compiler atoms are the  $E_8$  Casimir degrees [E] ([verification/v66\_e8\_casimir\_degrees.py])**

The load-bearing integers are not an ad-hoc list: they are the fundamental *invariant degrees* of  $E_8$  (exponents +1),

$$\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\},$$

with the clean (non-trivial) readings  $2=|\mathbb{Z}_2|$ ,  $8=\text{rank } E_8$ ,  $14=\dim G_2$  (the  $[K, Q]$  commutator  $G_2$ , flavor sector),  $20=\det L$ ,  $24=|W(A_3)|=4!$ ,  $30=h(E_8)$  (and the fittable  $12=|R(A_3)|$ ,  $18=p_4(a)=2+2^4$ ). Two exact sums close it:

$$\sum \deg(E_8) = 128 = 2^7 \quad (\text{the de Sitter prefactor } 128c_3^4; 7=\text{scalaron exp}),$$

$$\sum \exp(E_8) = 120 = |R^+(E_8)|.$$

Since  $E_8$  is the audit hull, its invariant degrees being the theory's integers is *structurally expected* — an organizing simplification that confirms the  $E_8$  choice rather than a coincidence.

### Two-level foundation, and the parameter audit

**Local foundation:**  $c_3, g_{\text{car}}$  are the two operational inputs (the two axioms). **Global bootstrap foundation:**  $g_{\text{car}}$  and the integer in  $c_3$  are  $E_8$ -closure fixed points; only  $\pi$  is primitive. The audit:

| former input                 | status now           | recovered by                                                                 |
|------------------------------|----------------------|------------------------------------------------------------------------------|
| $g_{\text{car}} = 5$         | discrete fixed point | rank-fill, Coxeter-match, integer-glue/norm, $h(E_8)$                        |
| 8 in $c_3$                   | discrete fixed point | $h(D_5)$ , rank $E_8$ , $\det R$ , $\varphi(30)$ , $2 \mu_4 $ (seam winding) |
| 6 in $\varphi_{\text{tree}}$ | discrete fixed point | $ R^+(A_3)  = h(E_8)/g_{\text{car}}$                                         |
| $\pi$                        | geometric primitive  | Möbius / Gauss–Bonnet ( <i>not derived</i> )                                 |
| $\varphi_0^{\text{ret}}$     | derived readout      | $\frac{4}{3}c_3 + 48c_3^4$                                                   |

**Correct one-liner:** *TFPT has no freely tunable fundamental numbers left; the only remaining continuous origin is  $\pi$  from the Möbius boundary geometry.* (Not “zero axioms”). [E]/[O]

## 17 The celestial and twistor continuum route

The previous section identified the seam's boundary 2-sphere with the celestial sphere of null directions ( $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ , clock = order-4 Möbius map [verification/v180\_clock\_is\_mobius.py]). Taking that identification seriously and reading the seam through the celestial-holography lens (the celestial-chiral-algebra construction of Bittleston–Homans–Sharma on ALE spaces; Costello's twistorial SDYM) produces a coherent thirty-step story — the narrative arc from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow — executed as work packages WP1–WP5d (both WP5d stages) plus the WP5e  $\alpha, \beta, \gamma, \delta_1, \delta_2, \varepsilon_2$  and  $\varepsilon_1$  stages, the three back-reaction milestones M1–M3, the measure decision (Step 21) and the constructive  $w_m$  derivation (Step 24) of the research contract CELEST.SEAM.01, plus the RP-blindness decision on the alignment bit (Step 22) with its twist-state completion (Step 26) with its twist-class definition (Step 29) and the interacting FK-toy scoreboard extension (Step 30), the thermal-seam third leg of  $c_3$  (Step 27) with the silver-eigenvalue demystification (Step 28), and the  $\alpha, \beta_1$  and  $\beta_2$  stages of the OS twistor bridge W0IT.OS.TWISTOR.01 (Steps 20, 23 and 25), whose full technical statement, kill tests and typing fence live in `tfpt_research_contracts`. Every step below is exact machine arithmetic [E] plus explicitly named identifications [C]; the route is a *programme* [O], not a claim, and nothing in it moves SEAM.EQUIV.01. The name of the route is deliberately *structural* (“celestial and twistor continuum route”, matching the contract



heading): what the route supplies is a candidate *continuum* construction on the twistor side, and the object diagram, the exact group-extension definition ( $\Gamma_{\text{ALE}}$  vs the normalising clock  $\langle \rho \rangle$ ) and the  $A_3$  role table that fence its semantics live in the contract (`tfpt_research_contracts`, `CELEST.SEAM.01/WOIT.OS.TWISTOR.01`).

**Step 1 — the setup: the  $\mu_4$  glue is a flat  $\mathbb{Z}_4$  monodromy on the  $A_3$  ALE space [E].** The  $E_8$   $\mu_4$ -glue grading  $240 = 52+64+60+64$  (v128) is *inner*:  $h = (2, 2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$  — a flat  $\mathbb{Z}_4$  monodromy in the Kronheimer–Nakajima sense on the  $A_3$  ALE orbifold  $\mathbb{C}^2/\mathbb{Z}_4$  (which *is* the  $A_3$  singularity  $XY = Z^4$ , exactly). The glue-equivariant  $\text{SDYM}(E_8)$  sector closes as a Lie algebra with graded dimensions  $60(d+1)/64(d+1)$  — possible *only* because  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$  — with zero modes = the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$ , and the four glue-sector characters sum *exactly* to the  $(E_8)_1$  character  $\chi = 1+248q+4124q^2+34752q^3$  (v377). The critical correction: the  $A_3$  deck induces on the celestial sphere only the sheet flip  $z \mapsto -z$ ; the order-4 clock  $z \mapsto iz$  is realised as the Kähler  $U(2)$  phase  $\text{diag}(i, 1)$ , and **clock**<sup>2</sup> = **deck** exactly (spin bridge  $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$  — the  $c_3 = 1/(8\pi)$  seam winding integer). [`verification/v492_celestial_z4_orbifold.py`] That spin bookkeeping is now *forced* on the seam fermions: the NS implementer of the deck satisfies  $U^2 = (-1)^F$  exactly (nonsplit  $\mathbb{Z}_4$  extension, no phase choice escapes it), and the quarter-shift lifts to a canonical  $\mathbb{Z}_8$  tower with  $V^2 \propto U$ ,  $V^4 \propto (-1)^F$  — “ $8 = 2|\mu_4|$ ” fermionically explained; the R-sector control splits ( $U_R^2 = +1$ ), so the forcing rests exactly on the established bounding spin structure. [`verification/v506_seam_clock_rigidity.py`]

**Step 2 — the deformation:  $a_0$  is a pure scale and the clock is the monodromy [E].** Clock-invariance selects from the versal  $A_3$  family exactly the one-parameter slice  $XY = Z^4 + a_0$ , and  $a_0$  carries *no* shape modulus: the binary-quartic invariants are  $I = 12a_0$ ,  $J = 0$  *identically*, so  $j = 1728$  and the  $\tau=i$  pillowcase (v168/v214) is frozen for every  $a_0$ . The clock acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (eigenvalues  $\{i, -1, -i\}$  = the  $A_3$  exponents,  $h(A_3)=4=|\mu_4|$ ) and *is* the Picard–Lefschetz monodromy of the family; the deformed-algebra pattern transfers  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$  grade-conserving (no  $\mu_4$ -sector leak). [`verification/v493_celestial_wp2_clock_deformation.py`] The converse is now tested *light*: the free field alone does *not* select  $\tau = i$  (the global  $\det'$  winner is the hexagonal torus, whose  $\mathbb{Z}_6$  carries no order-4 element) — the square is pinned by the clock’s *rigidity* (the raw DtN is mod-4 block-diagonal *iff*  $\tau = i$ ), and this order-4 clock is the *same* carrier datum as the P2 cusp class (v499): one common residual, not two ([`verification/v503_qgeo_emergence_light.py`]; the v215 deferred emergence test, light version — no marker moves, `QGeo.REALIZE.01` stays [C]/[O]). That common residual is now reduced to *one bit*: the deck  $z \mapsto -z$  has exactly two Möbius square roots ( $z \mapsto \pm iz$ ), both automatically of order 4, and a marks-preserving root exists *iff* the deck is the *central* involution of the mark-stabiliser  $D_4$  — given that alignment bit, order, uniqueness up to  $\text{Aut}(\mathbb{Z}_4)$ , the free 4-cycle and  $\tau = i$  (scale free) are all theorems; harmonicity alone is *not* enough on bare mark data (exact counterexample — though on free-deck circles it becomes sufficient, see below), and the fermionic  $\mathbb{Z}_4$  forces the *order*, not the alignment ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`; markers unchanged). The bit itself has now survived its tautology attack: marks and deck *do* share one torus origin — the marks are the four half-periods  $\Lambda/2\Lambda$  of the seam double, and *every* mark-free collar deck is a half-period translation (in all three stabiliser types the freely-acting involutions are exactly the three  $\sigma_c$ ), so the deck’s *class* is derived — yet the origin does not force the alignment: the marks-preserving root exists *iff* the deck pairs separate harmonically ( $\lambda \in \{-1, 2, \frac{1}{2}\}$ , one value per half-period), the aligned half-period is the CM-fixed one  $c^* = (1+\tau)/2$  (the clock’s curve lift needs the unit  $i$  of  $\mathbb{Z}[i]$ , while half-period decks lift rationally), and the silver counterexample sits *in-family* as “ $\tau = i$  with the deck a non-CM-fixed half-period”. The bit is thereby an equivalence *tetrad* — clock  $\iff (\tau=i \text{ and deck = CM-fixed half-period}) \iff (\text{Stab} = D_4 \text{ and deck central}) \iff \text{harmonic deck pairs} \iff \text{nonsplit NS Fock lift}$  — with a physical face: the central arrangement carries the nontrivial Fidkowski–Kitaev-type extension class ( $U^2 = (-1)^F$ ), the edge arrangement splits ( $U_e^2 = +1$ , zero roots, no  $\mathbb{Z}_8$  tower); only the *position* (which half-period) stays the carrier input ([`verification/v507_seam_bit_origin.py`], `SEAM.BIT.ORIGIN.01`; no marker moves). And that position half is now *topology*: the collar deck is

the deck transformation of a *covering* (the  $\mathbb{Z}_2$  seam of the Möbius double), and covering decks act freely — on the seam circle the unique free involution is the antipode ( $\text{Aut}(C_{16}) = D_{16}$ , complete census; every reflection has two circle fixed points, and its quotient is an *interval*, breaking the closed-circle  $6\pi$  budget), the NS Fock lift is nonsplit *iff* the deck is free (all 17 involutions in exact  $\text{Cl}(16)$ ), and the edge/silver arrangement — whose mark-fixing real structure has its fix circle *through* the deck poles — is the restriction of the *branched* pillowcase involution, not a covering deck: excluded. On free-deck circles “harmonic” already solves to  $b = \pm i$ , so the alignment bit reduces from “harmonic *and* central” to the square-modulus datum  $\tau = i$  alone — the position half is topology, the modulus half stays the one carrier input ([`verification/v510_seam_bit_freedom.py`], `SEAM.BIT.FREEDOM.01`; markers unchanged). The modulus half now has a sharp *discrete* face: on the free seam circle the deck-invariant 4-mark configurations form the one-parameter family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ , and *bare* mark-transitivity is automatic for every  $\delta$  (the pair-exchanging  $V_4$ ; all undirected per-mark data are identical, so no local jet sees the bit — “transitivity from v216-locality” is falsified) — what selects  $\delta = \pi/2 \iff \tau = i$  is exactly *flag* transitivity (marks *and* their two sides indistinguishable, the  $V_4 \rightarrow D_4$  symmetry lift), welded into a 13-fold exact equivalence web (clock rotation, mark mirror, odd-doublet split  $(2/\pi) \ln \cot(\delta/2)$ , arc-Laplacian degeneracy, the **v503** mod-4 indicator in closed form,  $\rho$ -twist existence, cusp-4-cycle implementability, harmonicity,  $j = 1728$ , and — since Step 22 — the  $D_4$  closure of the OS-reflection group); the carrier input thereby reduces from a continuous modulus to *one discrete symmetry-lift bit* whose negation is concretely measurable ([`verification/v512_seam_tau_flag.py`], `SEAM.TAU.FLAG.01`; markers unchanged).

**Step 3 — consistency, honestly demoted [E]/[C].**  $E_8$  is on Costello’s axionic-consistency list (no independent quartic Casimir; the Okubo coefficient is *derived*, not quoted), and the Green–Schwarz alignment is exact:  $\tilde{\lambda} = 6$  and  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$ . *But* the look-elsewhere battery shows the same alignment format passes for *all eight* algebras on the list (8/8): *alignment survives; selectivity does not* — the  $c_3$ -connection is compatibility plus genuine  $\lambda$ -arithmetic, not  $\mathfrak{e}_8$ -selective evidence. [`verification/v495_celestial_wp3_gs_alignment.py`]

**Step 4 — the type mismatch and the boundary limit [E].** The  $(E_8)_1$  character is *not* a conformal block of the celestial S-algebra in its own jet grading — the jet Fock grows as  $n^{2/3}$  against the character’s  $n^{1/2}$ , and the level-2 truncation has no jet analogue. But the boundary/period reading holds *exactly* at the current stratum: one full  $\mu_4$  period of loop energies sums to 248, and the glue-diagonal weights  $(0, 1, 1, 1)$  are integers. The character is a *boundary-limit shadow* of the S-algebra — precisely the scaling-limit shape of `SEAM.EQUIV.01` (MMST: the rational net exists in the limit of the lattice collar, not in the pre-limit algebra; **v336/v449**). [`verification/v496_celestial_wp4_character_shadow.py`]

**Step 5 — the null ideal, quantitatively [E].** The shadow is made a *precise coefficientwise limit*: the exact  $\chi_w$  family contains the jet grading at  $w=2$  and stabilises with threshold  $w = n+1$ . The deleted ideal is *derived* from root data (Freudenthal + Weyl + peeling):  $\text{Sym}^2(248) = 27000+3875+1$  with  $27000 = 30^3 = (h^\vee)^3$  exactly, and the level-2 quotient  $31124-27000 = 4124$  equals the independent  $\mu_4$  theta-split sector sum — two routes, one number. The  $SO(16)_1$  negative control separates: four blocks,  $5304 \neq 14^3$ , half-integer weight breaking one-block fusion. [`verification/v497_celestial_wp5a_null_ideal.py`]

**Step 6 — the deleting object as an operator [E].** The level-2 singular vector is constructed explicitly,  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ , in a machine-built Chevalley/Frenkel–Kac basis, and  $J_1^a|s\rangle = 0$  is machine-verified for all 248 generators (case tally 190/57/1 — exactly one case sees the level). The *level dial* is the  $F_1^\theta$  coefficient  $2(1-k)$ : only  $k=1$  deletes — without the central extension the deletion operator does not exist.  $\mu_4$  data: glue class  $j(\theta) = 1$ , clock phase  $-1$  on  $|s\rangle$ . The  $D_8$  contrast types the finding honestly: the singular-vector mechanism is level-1 *generic*; the *one-block closure* is the  $E_8/\mu_4$ -specific part. [`verification/v498_celestial_wp5b_singular_vector.py`]

**Step 7 — the limit state [E]/[C].** The state exists: the quasi-free family  $\omega_w$  (affine  $k=1$  vacuum on the currents +  $x^{wr}$ -contracted radial oscillators) stabilises exactly at the WP5a threshold  $w = N+1$ , and its limit carries the null ideal in its GNS kernel — the complete 9361-block exact

level-2 Gram has rank 4124 with kernel  $27000 = V(2\theta)$  weight by weight,  $|s\rangle$  is the GNS zero vector, and a CCR obstruction shows no  $w$ -uniform state could do this (the family is *necessary*).  
[verification/v500\_celestial\_wp5c\_gns\_limit\_state.py]

**Step 8 — the two-interval index [E]/[C].** The Haag-duality discriminator is measured on the seam lattice edge: the fermionic two-interval mutual information is extensive ( $\mu=1$  reference) while the orbifold prescription pays exactly one classical bit (the  $\ln 2$  plateau at machine precision), giving  $\mu_{\text{gauged}} = 4 =$  the v490 parity census, and the condensation arithmetic  $16 \rightarrow 4 \rightarrow 1$  (KLM/Longo–Rehren,  $\theta_v = 1$  at  $\nu = 16$ ) closes with  $\mu = 1$  — the preregistered kill (“ $\mu$ -offset  $\neq 0$  after condensation”) does not fire. [verification/v501\_celestial\_wp5d\_two\_interval\_index.py]

**Step 9 — the prefactor and the level, pinned on the CFT side [E]/[C].** The two exactly computable consistency anchors of the twistor uplift hold. The  $q^{-1/3}$  prefactor of  $E_4/\eta^8$  is exact  $\mu_4$  vacuum-energy bookkeeping: the clock is *inner*, so the twist on the 8 torus bosons is  $\theta = 0^8$  in every sector (a *shift* orbifold — not a rotation orbifold) and each sector carries the same  $-c/24 = -\frac{1}{3}$  at  $c = 8$ ; the sector weights  $(0, 1, 1, 1)$  are Casimir energies, via spectral flow and exactly via the 16-Majorana seam carrier (R–NS shift  $n/16$ :  $\frac{5}{8}, \frac{3}{8}, 1$ ); and  $k = 1$  is forced three independent ways (current condition  $h(J) = k$ , conformal embedding  $47(k-1)(k+266/47) = 0$ , central charge  $240(k-1) = 0$ ) plus the Step-6 singular-vector dial ( $31124 - 27000 = 4124$  at  $k = 1$  only). Honest sharpening: glue- $h$  integrality holds at every level and fixes nothing — the naive integrality route is retired; the deck rotation reading fails ( $\frac{3}{16} \neq \frac{3}{8}$ ). [verification/v502\_celestial\_wp5e\_prefactor\_level.py]

**Step 10 — the KLM completion on the lattice [E]/[C].** Complete rationality (Kawahigashi–Longo–Müger) has three ingredients — split property, strong additivity, finite two-interval  $\mu$ -index — and with Step 8 supplying the index, the two remaining legs are now witnessed for the same orbifold prescription. Strong additivity is *algebraically exact*: with the shared boundary Majorana,  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  exactly (GF2 spans full, matrix rank  $32/32$ ), while disjoint intervals give exactly index 2 — the Step-8  $\ln 2$  bit, localised at the split point; entropically the touching defect stays *bounded* by  $\ln 2$  with the Ising  $\frac{1}{4}$ -exponent approach while the preregistered  $U(1)/\text{Dirac}$  control *diverges* (Klich–Levitov pinned): bounded-vs-divergent is the lattice discriminator (the naive “defect  $\rightarrow 0$ ” expectation is false at a sharp lattice split — bounded  $\Leftrightarrow$  finite index, Longo–Xu). The split property is quantitative: the coupling singular values fall at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0%, and the orbifold *inherits* the same ladder exactly ( $C \rightarrow -C$ ; the  $\Lambda^2 C$  compound — Longo heredity). The Pimsner–Popa bound is a one-line identity,  $E(a) - a/2 = PaP/2$ , attained in exact integers ( $\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]$ ,  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ), and the index is pinned by two independent routes ( $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{pp}}$ ). All three KLM ingredients now carry lattice witnesses; the continuum uplift is honestly fenced — “finite-group orbifolds of completely rational nets are completely rational” is *Xu’s theorem*, cited, not claimed. [verification/v504\_celestial\_wp5d\_klm\_completeness.py]

**Step 11 — the equivariant anomaly ledger on twistor space [E]/[C].** The twistor side of the uplift is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $\mu_4$ -glue monodromy is exact — denominators  $(2, 4, 2)$  with Dedekind sum  $5/4 = (|\mathbb{Z}_4|^2 - 1)/12$ , equivariant characters  $(248, 0, -8, 0)$  by two routes, invariant average  $60 =$  the carrier. The honest sharpening: only the *invariant* sector is Okubo-quadratic ( $36\langle x, x \rangle^2$ ,  $36 = \tilde{\lambda}_{e_8}^2$ ); the twisted sectors are not, and the AB-weighted sum cancels the  $D_5$  quartic exactly while leaving the *rigid* residual  $32 T_3$  — twisted-sector closed strings must carry the rest. The *index bridge* is the centrepiece:  $f(m) = \text{ch}_2(T_m)$  exactly (fixed-point ledger = McKay/Kronheimer intersection ledger), with the integral glue defect  $-78$  by both routes; the level dials say  $k = 1$  *geometrically* (lattice current count  $240/0/0/0$ , embedding residual  $(0, 360, 814, 1362)$ , one scale  $\Rightarrow$  one level — since Step 13 a *theorem*, no longer a heuristic). And an honest *refutation*: the clock-invariant modulus  $a_0$  fills the BSS graviton slot  $O(2)$ , *not* the axion slot  $O(-2)$  (weight mismatch  $4 = |\mu_4|$ ) — “the theory brings its own GS axion as  $a_0$ ” is false; instead the three  $H^2(\text{ALE})$  classes carry exactly the three twisted-sector Coxeter characters  $\{i, -1, -i\}$  (the three twisted axion slots), and the bulk axion must come from the  $O(-2)$  tower field itself. The preregistered kill (“inflow demands a level  $\neq 1$ ”) does *not* fire on the equivariant skeleton.

[verification/v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py]

**Step 12 — the exchange no-go and the contact-term criterion [E]/[C].** Could the three sphere axioms of Step 11 cancel the rigid  $32 T_3$  residual by Green–Schwarz-type *exchange* with sector-compatible quadratic vertices? No — and the refusal is a theorem, not a fit. The  $W(D_5) \times W(A_3)$ -invariant vertex space on the glue Cartan is computed exactly (quadratics =  $\text{span}\{s_5, s_3\}$ , dim 2; quartics =  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$ , dim 5 — complete by Weyl nullspace arithmetic over all bidegrees), and the *product theorem* kills the mechanism wholesale: the product of any two invariant quadratics has identically zero  $T_5/T_3$  content, so *every* exchange image — any number of axioms, any selection rule, any couplings — is annihilated by  $\Phi_{T_3}$  while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . The enumeration sharpens where it fails: the strict two-index  $\mathbb{Z}_4$  rule collapses the sphere couplings entirely (only  $m = 0$  survives); the generous twist-insertion channels give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions, the second driven by the side discovery  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are parallel). Naturalness *dissolves*:  $\text{ch}_2$ -natural and Atiyah–Bott-weight couplings both carry certificate  $(0, 0)$  against the required  $(32, 72)$  — the ansatz space misses the target identically, scale- and coupling-independently. Controls:  $SO(16)$  has *no* sphere partners and uncanceled  $T_5$  ( $E_8$  is doubly special);  $D_8$  has no  $T_3$  structure at all. The slot bijection of Step 11 is untouched, and the preregistered level-kill still does not fire — what dies is the exchange *realisation* of the pairing; the  $32 T_3$  burden moves to the twisted BCOV contact terms, with a *binary* criterion: the one-loop coefficient on  $\mathbb{C}^2/\mathbb{Z}_4$  must come out exactly  $(9, -30, -15, 0, 32)$ , computed, not fitted.

[verification/v508\_celestial\_wp5e\_gamma\_pairing.py]

**Step 13 — one level is a theorem; the sector counter pins  $k = 1$  [E]/[C].** The Costello–Paquette–Sharma level-from-flux mechanism (Burns holography: “level = flux quantum  $\times$  Dynkin index of the boundary matter”) is executed on the lockstep spheres. The CPS skeleton is replicated exactly ( $S^3$  period  $(2\pi i)^2 N$ ; exceptional-sphere Kähler flux  $2\pi N$ ; boundary level magnitude  $2N$  by the symplectic-boson contraction,  $T(\text{vector}) = 1$ ), and the geometric side is *pinned*, *not postulated*: the Step-11  $\text{ch}_2$  ledger + integrality + unimodularity + effectivity force the pairing matrix  $c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by complete enumeration (48 unimodular solutions, exactly 2 effective: identity + diagram flip), so the glue fluxes are  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  with *exactly one* quantum per charged root current. The consistency test then does real work: “level = total open-string flux” is *killed by the lockstep test itself* (three unequal numbers —  $F_i$  is the flavour multiplicity), while “level = flux per adjoint current” gives  $(1, 1, 1)$ , anchored by the lattice current count  $(240, 0, 0, 0)$  and the embedding index 1. The ord-4-vs-level-1 tension resolves — the clock order counts *fractional flux sectors* ( $\mu = 16 = \text{ord}^2$ , Lagrangian  $\mu_4$  glue, KLM  $16 \rightarrow 1$ ) — and a new dial pins the value:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$ , exactly *one* sector iff  $k = 1$ . The centrepiece is the *lockstep theorem*, lifting Step 11’s “one scale  $\Rightarrow$  one level” from heuristic to theorem: clock invariance forces the four branch points into one  $\mu_4$  orbit, the period moduli  $|\Pi_j| = \sqrt{2}t$  are equal on all three spheres, and  $\det(A - 1) = -4$  leaves no invariant flux vector —  $k_1 = k_2 = k_3$  *is a theorem* of clock invariance, with a new falsifier (the clock-forbidden family  $Z^4 - Z$ : zero lockstep chains over all 24 orderings) and an honest limit (uniform doubling keeps the lockstep: the theorem proves *equality*, the value 1 comes from the counters). Controls:  $k = 2$  dies on the closure dials while the lockstep dial honestly does not fire;  $SO(16)$  leaves two spheres flux-dark;  $A_2$  admits no Lagrangian glue at all ( $\mu = 12$  not a square). The CPS dictionary on  $\mathbb{P}^1/\mathbb{Z}_4$  itself stays [C] (their geometry is the  $\mathbb{C}^2$  blow-up, not the  $A_3$  ALE); the type-I B-model back-reaction stays [O].

[verification/v509\_celestial\_wp5e\_eps2\_level\_from\_flux.py]

**Step 14 — the full-tensor ledger: the collapse is real, and one cubic door opens [E]/[C].** Step 12’s collapse was a *Cartan-restricted* statement — the named escape route ( $\delta_2$ ) asks whether the full adjoint tensor structure of  $\mathfrak{e}_8$  (non-Cartan external legs, higher-arity vertices) reopens the pairing. The answer has two halves, both exact. *The collapse is confirmed*:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  is semisimple with no  $\mathfrak{u}(1)$  factor (class-0 roots  $40+12$ , root span rank 8), the sectors factorise into minuscule Weyl orbits  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$ , and the *innerness theorem* decides everything wholesale: the grading element  $h$  lies in the Cartan of  $\mathfrak{g}_0$ , so every  $\mathfrak{g}_0$ -invariant



tensor carries total sector charge  $0 \bmod 4$  — bilinear invariants exist only at  $j' = -j$  (dims  $2/1/1/1$ ) and *all* 15 non-neutral trilinear sector triples have  $\text{Hom} = 0$ : the sphere axions have no invariant partner operator at any arity  $\leq 3$ , full-tensorially. *And one cubic door opens*: of the nine neutral trilinear structures (dims  $(3, 2, 2, 1, 1)$ ) all cross-sector ones are pure brackets, but the *unique* totally symmetric survivor — the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir, so  $T_5$  stays protected) — carries an exchange quartic  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  (two independent routes, Killing propagator  $2h^\vee = 60$  from the roots) with  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ : Step 12's master kill covers *quadratic* vertices only and does not extend to cubic ones. The pairing nevertheless remains obstructed in the charge reading —  $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  has rank 3 against augmented rank 4 — but with the genuinely *weaker* certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$ ; relaxed, it becomes *exactly solvable*:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$ . Number fences typed **[C]** only:  $64 = \dim \mathfrak{g}_1 = 2^6$ ,  $1920 = |W(D_5)| = 8 \cdot 240$ . Controls:  $SO(16)/D_8$  has no symmetric cubic, no odd sectors and no  $A_3$  block ( $E_8/\mu_4$  doubly special); false  $\mathfrak{g}_0$ /sector assignments inflate or kill the tables. The burden on the  $\delta_1$  contact terms is thereby *sharpened*: they must supply either the  $\psi = 64$  slice or the selection-rule relaxation — a burden since *discharged* by Step 17 (the declared KS measure supplies the  $\psi = 64$  slice exactly, so the cubic  $d$ -channel is not needed) — and the  $\delta_1$  run, undecided at exploration level at the time (the strict holomorphic reading is refuted at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor — a method boundary, not a kill; the contact term is the *modular completion* of the Atiyah–Bott data, a Harvey–Moore-type  $\tau$ -integral with the forced leading  $(T_5, T_3)$  ratio 4:3), has since been *decided* in Step 19 (kill under the derived measure, v518). [\[verification/v511\\_celestial\\_wp5e\\_delta2\\_tensor\\_ledger.py\]](#) The 1920 fence itself has since been stress-tested and *fails as a derivation claim*: the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded (11/924 catalog expressions hit 1920, vs 8/924 for the control target 1800) and convention-contingent (only 1 of 5 normalisations produces 1920; charge-legal flux pairings give 2048/1800 and miss; the twisted cubic index  $32i$  clashes in class provenance with the true quartic 32) — the convention-stable **[E]** core is the factorisation  $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$ , the fence stays **[C]**, and the physical generation of the 32 stays **[O]** ([\[verification/v513\\_celestial\\_dterm\\_nonderivation.py\]](#), CELEST.DTERM.NONDERIV.01; no marker moves).

**Step 15 — the bulk axion is a construction;  $\tilde{\lambda} = 6$  is pinned three ways **[E]**/**[C]**.** Step 11 refuted “ $a_0$  is the GS axion” and left the bulk axion to the  $O(-2)$  tower field *as a slot*; the  $\varepsilon_1$  stage now *builds* it. On  $\mathbb{P}T' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)$  the pushforward ledger  $\pi_* O(-2)$  closes the Penrose accounting exactly ( $(d+1)^2$  classes per degree = the exact wave-operator nullspaces,  $d \leq 6$ ), and *equivariantly*: the clock acts on the fibre coordinates as  $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$ , the incidence relation forces the column weights  $(+1, +1, -1, -1)$ , and the bookkeeping closes *block by block* for all four characters. The character series are  $P_0 = 1 + 3t^2 + 15t^4 + \dots$ ,  $P_1 = P_3 = 2t + 8t^3 + \dots$ ,  $P_2 = 6t^2 + 10t^4 + \dots$ ; the  $d = 0$  slot has multiplicity  $(1, 0, 0, 0)$  — *the bulk axion survives the projection* — and the invariant fibre ring is the *hypersurface*  $(1 - t^8)/((1 - t^4)^2(1 - t^2))$ :  $Z \in O(2)$ ,  $X, Y \in O(4)$ , one relation in degree 8 = the  $a_0$  weight (the Step-1/Step-2 geometry re-emerging from the slot); the Step-11 bijection sharpens to a per-character statement (twisted minimal content  $(2t, 6t^2, 2t) =$  the Coxeter eigenvalues, no degree-0 mode,  $\det(A - 1) = -4$ ), and the graviton control separates  $a_0$  cohomologically (the  $O(+2)$  slot starts invariant only at fibre degree 4 with multiplicity  $3 = \{X, Z^2, Y\}$ ). The Green–Schwarz residue is then pinned *three* independent ways: Okubo ( $\text{Tr}_{\text{adj}} X^4 = (6\langle x, x \rangle)^2$  exactly on the 240 glue roots,  $36 = h^\vee + 6$ ,  $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\text{fam}}$ ), the measure chain on  $\mathbb{P}T/\mathbb{Z}_4$  (the quotient factor  $\mu = 1/4$  enters vertex<sup>2</sup>/propagator/anomaly and cancels *exactly*; the wrong bookings are quantified and excluded: anomaly-only  $\Rightarrow \tilde{\lambda} = 3$ , missing propagator renormalisation  $\Rightarrow \tilde{\lambda} = 12$ ), and the flux side (minimal CPS quantum  $N = 1$ , a single exchange channel exactly at  $k = 1$ ,  $(\kappa/c_3)^2 = 12$ ). The first honest *back-reaction* step follows in Gibbons–Hawking form: clock-invariant four-centre configurations form exactly two branches (four axis points = pure resolution, one free  $\mu_4$  orbit = pure deformation), the free orbit re-derives the Step-2 family ( $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$ ,  $a_0 = -z_0^4$ , monodromy = one clock step), the charge-4 GH point is exactly the  $S^3/\mathbb{Z}_4$  seam boundary, the periods  $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$  confirm the Step-13 lockstep *from geometry* ( $\Pi \cdot A = i\Pi$ ), the source ledger carries charge  $4 = |\mu_4|$ , and the

honest sharpening: the Ricci-flat ALE has asymptotic log coefficient *exactly zero* (the CPS log is an exceptional-locus statement; Burns contrast  $\det g \neq 1$ ), with the multipole selection rule  $m \equiv 0 \pmod 4$  storing  $-a_0$  in the first symmetry-breaking  $(4, \pm 4)$  harmonic. Controls: **so<sub>8</sub>** ( $\tilde{\lambda}^2 = 12$  irrational — perfect squares in the Deligne series only  $\{9, 36\} = \{\mathfrak{sl}_3, \mathfrak{e}_8\}$ ),  $\text{diag}(i, i)$  (Veronese cone, no hypersurface, the bijection collapses),  $k = 2$  (three exchange channels). What stays honest: the chain is conditional on Costello’s flat- $\mathbb{P}^1$  matching [C], and the *quantised* BCOV coefficient on  $\mathbb{P}^1/\mathbb{Z}_4$  plus the twisted channels ( $32 T_3$ ) stay [O] — preregistered as the M1–M3 back-reaction milestones (the  $A_3$   $\Omega_N$ , the twisted KS measure, the  $a_0$  uplift), each with success and kill criteria; all three are executed in Steps 16–18. [verification/v514\_celestial\_wp5e\_eps1\_axion\_slot.py]

**Step 16 — the back-reacted  $\Omega_N$ : closed form, integral periods, forced charge 4 [E]/[C].** The M1 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS criterion met and neither KILL fired. The residue 2-form of the family is *derived*, not assumed: on  $F = XY - P(Z)$  all three ambient representatives satisfy  $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$  and pull back to  $\omega_2 = dX \wedge dZ/X$ ; at  $a_0 = 0$  the orbifold-cover pullback is  $\omega_2 = 4 dz_1 \wedge dz_2 = |\mu_4| \times (\text{flat form})$  — the residue normalisation itself *carries* the source charge 4 — and the clock multiplies  $\omega_2$  by  $i$  (the Step-1  $\det = i$  replicated). The four  $O(2)$  centre sections  $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$  close the twistor family in one line,  $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$  ( $e_1 = e_3 = 0$  identically; the seam fibre  $\lambda = 0$  is exactly the Step-2 family, the  $a_2 \in O(4)$  slot opens only off-seam), with the CY-compatible clock lift  $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$ ,  $\gamma^4 = 1$ ,  $\Omega \rightarrow +\Omega$ . The periods reduce generically ( $\int_{S_j} \omega_2 = 2\pi i(q_{j+1} - q_j)$  for arbitrary profile and path), giving the seam-fibre lockstep vector  $2\pi i t_0(i-1)(1, i, i^2)$  with the exact covariance  $\Pi_{j+1}(-i\lambda) = i \Pi_j(\lambda)$  for *all*  $\lambda$ , and the 12 collision nodes — honest conifold points (Hessian  $\det = 512 t_0^6$ , quadric rank 4) — sit exactly on the 8 eighth roots of unity ( $8 = 2|\mu_4|$ , the Step-1  $\mathbb{Z}_8$  bridge), in clock orbits  $4+4+4$ . The back-reacted 3-form is then *closed form*:  $\Omega_N = \Omega_0 + \sum_p N_p K_p$  with the CPS/Bochner–Martinelli kernel on the four centre twistor lines ( $\int_{S^3} K = (2\pi i)^2$  exactly,  $dK = 0$  off-source, scale degree 0); the undeformed  $\Omega_0$  carries *zero* quantised 3-flux (all flux is sourced), every 3-cycle period is  $(2\pi i)^2$ -integral, the flux vector  $N(1, 1, 1, 1)$  is forced uniform *twice over* (clock orbit + the  $K_4$  connectivity of the four lines), and the lens fundamental domain *forces*  $N \in 4\mathbb{Z}$ : the source charge  $4 = |\mu_4|$  from quotient large-gauge quantisation, minimal invariant charge  $\leftrightarrow$  the CPS quantum  $N = 1 \leftrightarrow k = 1$  (Step 13). The honest fence has teeth: on the clock-forbidden family  $Z^4 - Z$  the  $(2\pi i)^2$  quantisation *also* holds — integrality alone is *not* the discriminator; the discriminator is the lockstep phase/modulus structure (0/24 orderings vs 8/24, node support on twelfth roots instead of  $\mathbb{Z}_8$ ) together with the clock forcing of equal fluxes. Anchors and controls:  $\text{EH}/\mathbb{Z}_2$  forces charge  $2 = |\mathbb{Z}_2|$  from the same machinery, the CPS flat patch allows every integer  $N$  (the unconstrained anchor),  $\text{diag}(i, i)$  collapses at step zero ( $\Omega_0 \rightarrow -\Omega_0$ ), and fractional charges  $N = 1, 2, 3$  are excluded. Global kernel patching, the Hitchin small resolution and the CPS brane dictionary stay [C]; M2–M3 are executed in Steps 17–18, and the quantised BCOV coefficient stays [O] only in its measure question (the  $\delta_1$  chain itself is decided in Step 19). [verification/v515\_celestial\_wp5e\_m1\_omega\_n.py]

**Step 17 — the twisted KS measure: every sector the same Okubo square [E]/[C].** The M2 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording met *on the declared completion measure* and the KILL (a leftover independent quartic in *any* sector) not fired. The measure ansatz is declared before computing, and honestly typed: Step 12 killed every *exchange* realisation, and the  $\delta_1$  exploration identified the twisted contact term as the *modular completion* of the very Atiyah–Bott data that produced  $A_{\text{fix}}$ ; M2 declares the completion contact term per twisted box channel,  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  — in the channel with the  $g^j$  zero-mode normalisation  $1/\det_j$ , the twisted-sector *loop* contributes the *unphased* sector trace, of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part. Everything downstream of the declaration is exact arithmetic, and it locks in place with no dial to turn: per glue class the contact weights obey the *completion-weight identity*  $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$  ( $h_m = m(4-m)/8$  the Step-9 sector Casimir energies,  $\text{ch}_2(T_m)$  the McKay/Kronheimer charges of the Step-11 index bridge): the three sphere axions pair through their *own*  $\text{ch}_2$  charges — an exact identity, *no free scale, no fit*. The locks are parameter-free:  $T_5 = 0$  for *any* scale  $c$

(the  $\text{ch}_2$  pattern alone), the ratio  $h_2:h_1 = 4:3$  reproduces the  $\delta_1$ -forced leading ratio, and the  $T_3$  budget fixes  $c = 4 = |\mu_4|$  uniquely (leading system  $\det 256$ ). The success table is then uniform:  $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x \rangle^2/\det_j$  — a *perfect Okubo square* ( $T_5 = T_3 = 0$ ,  $\tilde{\lambda}^2 = 36$ ) in *every* twisted channel; the total  $A_{\text{fix}} + \text{contact} = 45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo}$  is exactly the *unique* quartic-free weighting of the Step-11 rigidity theorem. Both Step-12 certificates are killed ( $\Phi_{T_3} : 32 \rightarrow 0$ ,  $\Phi_P : 72 \rightarrow 0$ ), and the Step-14 slice  $\psi = 64$  is *supplied exactly* — the cubic  $d$ -channel is *not needed* ( $c_d \text{ free} = 0$ ). The remainder sits inside the exchange span (the bulk axion cancels every channel with the *same*  $\tilde{\lambda}^2 = 36$ , the twisted propagator carrying the zero-mode factor  $1/\det_j$ ), and the Step-15 measure chain stays  $\mu$ -blind. Controls with teeth: wrong scale  $c \in \{1, 2, 3, 5\}$  leaves  $T_3 = (24, 16, 8, -8)$ ; the  $h_1 \leftrightarrow h_2$  shuffle breaks  $T_5$  and  $T_3$ ;  $SO(16)$  keeps  $T_5 = 20/T_3 = -40$  — the KILL *fires* there ( $E_8$  doubly special);  $\text{diag}(i, i)$  degenerates to the  $\mathbb{Z}_2$  target; and the  $\mathbb{Z}_2/\text{Eguchi-Hanson}$  anchor *passes* with the same mechanism ( $9\langle x, x \rangle^2 = \frac{1}{4} \times 36\langle x, x \rangle^2$ , coupling scale  $2 = |\mathbb{Z}_2| = \text{centre count}$ ). The mandatory fence: the completion reading (loop = unphased sector trace with the same zero-mode normalisation) is *declared* within this step, supported by the  $\delta_1$  modular-completion finding, *not* derived from the BCOV integral here — the  $\delta_1$  chain has since been decided in Step 19 (v518: the *derived* chiral measure fails all three testers), and the declared-vs-derived measure question has since been decided in Step 21 (v520: the completion reading is now *derived at probe level* from two independent constructive sources — F-independence + the Quillen pairing — under the typed premises TP-1..TP-4); the residual **[O]** is the constructive BCOV derivation of the  $w_m$  normalisation itself. [\[verification/v516\\_celestial\\_wp5e\\_m2\\_twisted\\_ks\\_measure.py\]](#)

**Step 18 — the  $a_0$  uplift: four coupled centre-count scales **[E]/[C]**.** The M3 milestone preregistered in Step 15 is now *executed*, with the preregistered SUCCESS wording (“a log-type correction whose coefficient is tied to the source charge  $4 = |\mu_4|$ ”) met and the KILL (decoupling from the centre count) not fired. The uplift object is the generalized-Legendre-transform kernel per centre,  $G(\eta_p) = \eta_p \log \eta_p$  (the Lindström–Roček / Ivanov–Roček dictionary, typed **[C]**); its derivative kernel  $\chi = \sum_p \log \eta_p = \log P_4$  is the *log of the Step-16 family polynomial*. The bridge is well-typed and exact: the  $O(2)$  section is a *null* coordinate ( $\Delta_{3d} G(\eta_p) = 0$  identically for an *arbitrary* kernel), and the residue identity  $\partial_x [\text{transform}(\log \eta_p)] = 1/r_p$  holds exactly — the Gibbons–Hawking potential  $V = \sum_p 1/r_p$  is the  $x$ -derivative of the contour transform of  $\chi$  (flux  $-4\pi$  per centre, source charge  $4 = |\mu_4|$ ). The  $a_0$  correction then arrives on *four coupled centre-count scales*: (i) the asymptotic kernel  $\log \chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$  carries the log coefficient  $4 = |\mu_4| = \text{centre count}$  (the CPS “ $N \log$ ” analogue at kernel level); at the clock-fixed seam fibre the  $a_2$  slot closes and the first correction is *exactly*  $a_0/\eta^4 =$  the  $(4, \pm 4)$  multipole in kernel clothes, with exact  $m$ -grading (the  $m = \pm 4$  pieces are *pure*  $a_0$ , the  $m = 0$  piece is  $\varphi_0$ -free) and *no*  $\log \times \text{power}$  terms — the Step-15 honest zero is preserved; (ii) the once-integrated GLT tower  $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + \dots$  obeys the selection rule  $p_n = 0$  unless  $n \equiv 0 \pmod 4$  with  $p_{4k} = 4(-a_0)^k$  — every correction carries the prefactor 4; (iii) the exceptional-locus  $\log \chi(0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$  ( $t_0$ -coefficient  $4 = |\mu_4|$ , clock phase  $4i$ ; the modulus response equals 1 *at* the locus and falls off as a power law at infinity — the log lives where Step 15 said it must); (iv) the period response is  $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$  *uniformly* (the periods are linear in  $t_0$ : lockstep and the phase ratios  $(1, i, i^2)$  are preserved exactly at first order), and integrating the shift around the  $a_0$  circle gives the monodromy  $e^{2\pi i/4} = i = \text{one Coxeter clock step}$  — the Step-2 monodromy reproduced from perturbation theory; the  $\mathbb{Z}_8$  node support is  $a_0$ -rigid, and the  $(2\pi i)^2$  fluxes are topological (they cannot move). Controls: the  $(4, 0)$  multipole is clock-invariant and breaks nothing; the  $\mathbb{Z}_2/\text{Eguchi-Hanson}$  analogue reads  $2 = |\mathbb{Z}_2|$  on *every* dial (amplitude, kernel log, exceptional log, period response  $1/2$ , monodromy  $-1$ ); wrong centre counts  $k = 3, 5$  move the coefficient to  $k$  with modulus slots  $O(6)/O(10) \neq O(8)$  — the observable *moves with the centre count*; the clock-forbidden family fails both dials ( $p_3 = 3 \neq 0$  breaks the selection rule,  $e_4 = 0$  kills the  $a_0$ -log). What stays **[O]**: the full nonlinear Kähler potential of the resolved  $A_3$  ALE (no closed form exists). [\[verification/v517\\_celestial\\_wp5e\\_m3\\_a0\\_uplift.py\]](#)

**Step 19 — the derived measure decides — and disagrees with the declared one**

[E]/[C]. The  $\delta_1$  question that Step 17 left open (*derive* the completion reading from the Harvey–Moore/BCOV  $\tau$ -integral) has now been executed as a three-stage probe chain and consolidated: instead of declaring or scanning a measure, the measure is *solved for* from blockwise  $SL(2, \mathbb{Z})$  covariance of the dressed 16-component Weil system. Four exact results. (i) *The completion closes*: the discriminant module of  $D_5 \oplus A_3$  is  $\mathbb{Z}_4 \times \mathbb{Z}_4$  with  $q = (5x^2 + 3y^2)/8$ ; the Gauss sums are  $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$  (signature 0 mod 8 = rank  $E_8$ ), the Weil relations hold exactly in  $\mathbb{Q}(\zeta_8)$ , the diagonal and anti-diagonal Lagrangians are the two  $E_8$  gluings, and the 16-vector theta  $S$ -covariance drops the naive 4-character-rule residual from  $2.91 = O(1)$  to  $\sim 10^{-39}$ . (ii) *The obstruction is a finite  $\mu_4$  character, identified*: the dressed gauge blocks carry a multiplier system whose  $SL(2, \mathbb{Z})$  relation defects are  $(1, 1, 1)$  exactly on all 15 sector pairs — a *character* of the orbit stabilisers  $\Gamma_1(4)/\Gamma_0(2)^\pm$ , not a genuine 2-cocycle — and on  $\Gamma_1(4)$  it is  $\lambda(\gamma) = i^{2B+C/4}$ . (iii) *The cancellation exists and is the twisted fibre block*:  $G[a, b] = f_1 f_3$  is an exact identity (the twisted  $\mathbb{C}^2$  gauge block is the weight- $(1, 3)$  axion pair), the T-fix mechanism is one line of exact phase arithmetic ( $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$ ), and the exact cancellation table reads: bare  $\rightarrow$  order 4, the three sphere axions  $f_1 f_2 f_3 \rightarrow 4$ ,  $f_2 \rightarrow 6$ ,  $f_1 f_3 \rightarrow 1$  — *only* the twistor-fibre content cancels the  $\mu_4$  system, with the strict solutions exactly  $\chi_4$  (dims  $(3, 3)$ ) and  $\chi_{10}$  (dims  $(1, 1)$ ), certified on the dressed functions. (iv) *All three preregistered testers fail*: under both derived solutions the integral misses  $T_5 = 0$  (fractions 0.5/0.83), misses the forced 4:3 leading ratio ( $W_{13} = 0$ ) and misses  $-A_{\text{fix}}/\text{the } \psi = -64N$  slice (spreads 1.05/1.47), and the honest  $(N_1, N_2)$  orbit rebalance rescues nothing in the positive cone — a genuine KILL on the derived surface. Controls: the  $\mathbb{Z}_2/\text{Eguchi–Hanson}$  anchor (residual order 2, cancelled to 1 by the same mechanism),  $SO(16)$  (the order-4 supply structurally absent), wrong form/wrong signature (break exactly). Fences [C]: the  $f_1 f_3 = \text{KS-weight identification}$  (the deck weights  $(1, 3)$  are the twistor fibre weights forced by the Step-15 incidence ledger — an exact match, *not* a complete BCOV derivation) and the kernel-family convention. TENSION, stated honestly: the *declared* completion reading (Step 17) delivers the  $\psi = 64$  slice and cancels the  $32 T_3$ ; the *derived* chiral measure (this step) fails all three testers — both are exact; the sharp open question was which of the two is the true BCOV measure. Neither result is hidden behind the other — and Step 21 has since decided the question at probe level in favour of the declared reading, sharpening this kill. [verification/v518\_celestial\_wp5e\_delta1\_derived\_kill.py]

**Step 20 — the real structure exists — and free reflection positivity picks the same family [E]/[O].** The  $\alpha$  stage of the OS twistor bridge WOIT.OS.TWISTOR.01 is executed (WOIT.THETA.FREE.01). The classification is complete: exactly *two* families of anti-linear structures on  $\mathbb{C}^2$  normalise the clock  $\rho = \text{diag}(i, 1)$  — family D ( $z \mapsto \mu \bar{z}$ ,  $|\mu| = 1$ ) satisfies  $\Theta \rho \Theta = \rho^{-1}$  *exactly* with  $\Theta^2 = +1$  ( $-1$  is *impossible*:  $M\bar{M} = \text{diag}(|\mu|^2, 1)$  — the clock-inverting family is Kramers-free), inverts the deck, and reflects the seam circle with two cut points; family A ( $z \mapsto \mu/\bar{z}$ ) centralises the clock projectively and *never* inverts it, with  $\Theta^2 = \pm 1$  per  $\mu$ . The role separation is sharp: family A *defines* the euclidean section — Woit’s  $\rho_{\text{tw}}$  ( $\rho_{\text{tw}}^2 = -1$ , no real points) is replicated exactly on  $\mathbb{C}^4$  and is family A globally — while family D *reflects* it (the OS conjugation;  $\sigma_{\text{std}}$  with real points  $\mathbb{RP}^3$ ). Mark compatibility pins  $\mu \in \mu_4$  (a 4-element torsor with  $\rho \Theta_\mu \rho^{-1} = \Theta_{-\mu}$ : two clock orbits); the  $\mathbb{Z}_8$  spin plane has no phase leaks ( $\zeta_8^k = \zeta_8^{-k}$ ); and in exact  $\text{Cl}(16)$  the Fock implementer  $\Theta_{\text{Fock}} = U_r \circ K$  has  $\Theta_{\text{Fock}}^2 = 2^7 > 0$  (normalised  $+1$ ), inverts the Fock clock tower ( $V \mapsto 4096 V^{-1}$ , exact scalar) and normalises the deck — while the *deck-induced* candidate has  $\Theta_t^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F$ : the Step-2/v510 split/nonsplit dichotomy *is* the  $\Theta^2 = +1$  vs  $(-1)^F$  dichotomy, so the deck does *not* furnish the OS  $\Theta$  — the seam-circle *reflection* does. Free reflection positivity then holds for exactly that pinned  $\Theta$ : on the bond cut (marks at the bond midpoints of the 16-Majorana circle) the one-particle Gram is positive definite  $((8, 0, 0)$ , min eigenvalue  $1.888 \times 10^{-3}$  at 40 digits), the even  $\deg \leq 2$  sector is  $(29, 0, 0)$ , and at  $N = 8$  the *complete* half-sided algebra is  $\text{RP}$  with no degree truncation; the twist  $\eta = +i$  is forced ( $\eta = 1$  non-Hermitian); the cut *through* the sites fails exactly ( $\det = 0$ , inertia  $(3, 3, 1)$  — a lattice-placement artifact, the continuum Cauchy–Stieltjes control is strictly positive), and the clock-centralising family-A structure fails  $\text{RP}$  *structurally*  $((4, 4, 0))$ :  $\text{RP}$  and  $\Theta \rho \Theta = \rho^{-1}$  select the *same* family. Bonus: the anti-chiral state flips the odd sector to negative definite — an exact free-level shadow of kill test 3, living entirely in the fermionic sector. **Kill**



**test 1 of the contract does *not* fire at the free/equivariant level and stays live on the interacting algebra;** the interacting algebra, gauge-fixed RP, OS reconstruction, chirality and  $\mu_4$  incidence remain open contract work (the  $\beta/\gamma$  milestones named in the contract). No marker moves. [verification/v519\_woit\_theta\_rp\_free.py]

**Step 21 — the measure decision: single-valuedness is derived, the completion reading wins [E]/[C].** The declared-vs-derived tension that Steps 17 and 19 left as the named open face is now *decided at probe level* (ERFOLG-A on the preregistered decision layer; the  $\delta_{1e}/\delta_{1f}$  consolidation). The centrepiece is exact linear algebra: the joint invariant subspace  $\{v : \rho(T)v = v, \rho(S)v = v\}$  of the dressed 16-component Weil system has  $\dim_{\mathbb{Q}} = 8 = 4 \times 2$  and *every* kernel vector lies in the  $\mathbb{Q}(\zeta_8)$ -span of the two Lagrangian gluings  $\{e_H, e_{H'}\}$  — both with theta series  $E_4$ , both carrying the *unphased* sector trace  $Q^{(0)} = (36, 72, 36, 0, 0)$ . Single-valuedness of the physical one-loop lattice factor — Step 19’s typed premise — is then *derived* from two independent constructive sources rather than postulated. *Source 1 (the fundamental-domain lemma):* every strict chiral closure at physical weight carries a nontrivial character ( $\chi_4(T) = e(1/3)$ ,  $\chi_{10}(T) = e(5/6)$ , exact; zero strict trivial-character solutions across all eleven dressings); the canonical column-matched member’s total block sum obeys  $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$  *pointwise* (certificate  $3.9 \times 10^{-16}$ , nonvacuous  $|G| \geq 59.6$ ), and an exact change of variables turns fundamental-domain independence into  $\int = 0$  for *every* chiral integrand: the Step-19 route was never a well-defined nonvanishing moduli integral — the kill is *sharpened*, its unfolded values are convention bookkeeping. *Source 2 (the Quillen pairing):* BCOV’s  $F_1$  pairs  $\text{hol} \times \text{conj}(\text{hol})$ , and the doubled transports satisfy  $|\bar{t}\bar{t} - 1| < 8.3 \times 10^{-40}$  on all 15 pairs ( $|\chi_4|^2 = 1$  exactly, while the one-sided square  $\chi_4^2 = e(2/3) \neq 1$  keeps defects on 15/15 pairs); the doubled 256-dimensional system closes *strictly* at trivial character and physical weight (nullspaces  $(28, 17) \geq 11$ ), contains the unphased diagonal *and* the physical columns  $w_b \otimes \bar{w}_b$ , and the wrong pairing  $\text{hol} \otimes \text{hol}$  is *empty*  $(0, 0)$ . The consistency circle then closes: the forced unphased numerator, inserted into the Step-19 scaffold, reproduces the Step-17 Okubo squares *levelwise* ( $\sum_m V \propto \langle x, x \rangle^2$  exactly on every level  $n \leq 8$ ;  $J$ -weighted spreads  $\sim 5 \times 10^{-41}$ , zero negative cells) with  $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$  — the  $J$ -weighted mirror image of  $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$ . The declared reading also wins the two decision-layer discriminators: the physical AB column  $\sum_m i^{bm} e_{(m,m)}$  is contained *exactly* in the  $\chi_4$  family ( $\sigma \sim 1.8 \times 10^{-16}$ , one scalar per orbit — canonicalising the  $(N_1, N_2)$  freedom) yet the canonical member still fails every tester; and at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor only the declared reading hits ( $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $\psi(A_2) = -8$ , scale tooth  $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$  unique) while all three derived instantiations fail. Typed premises [C] (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — it must source  $\psi = 64$ ), TP-3 (the Step-19 kernel convention), TP-4 (the Quillen structure of BCOV  $F_1$ ). What stayed [O] — the constructive BCOV derivation of the  $w_m$  normalisation itself — is since executed by Step 24, narrowing the [O] to the global BCOV integral beyond the fibre zero-mode factor. No marker moves. [verification/v520\_celestial\_wp5e\_measure\_decision.py]

**Step 22 — free OS positivity does not see the bit: the eighth side-blind test [E]/[C].** Could the Step-20 machinery *derive* the Step-2 alignment bit? The hypothesis — free RP or  $\Theta$  existence breaks for  $\delta \neq \pi/2$  on the deformed mark family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$  — is now decided *negative*, exactly as preregistered (KILL branch). The census is exact and symbolic: for *every*  $\delta$  the two mark-swapping reflections  $\Theta_{\pm e^{i\delta}}$  exist (all 20 cut/mark incidence solvesets empty), while mark-*fixing* reflections exist iff  $\delta = \pi/2$  (solveset  $\cos \delta = 0$ , lost immediately at  $\pi/2 - \varepsilon$ ). Free RP is  $\delta$ -blind in the strongest sense: the bond-cut Gram inertias are  $(8, 0, 0)$  PD for all tested  $\delta$  with the full sorted spectra *identical to 40 digits* (deviation 0.0); the  $N = 16$  odd- $m$  failure is a lattice-placement artifact (at  $N = 32$  all  $m = 1.7$  have mark-compatible bond axes with PD inertia  $(16, 0, 0)$ ); the continuum OS kernel in axis-adapted coordinates is *exactly*  $1/\sin((s+t)/2)$  — the axis position drops out identically — and the v510/v512 counterwitness passes free RP.  $\Theta$  existence is  $\delta$ -blind too ( $\Theta^2 = +1$ , deck normalisation, Fock implementability  $U^2 = +2^7/2^8 > 0$  for every  $\delta$ ); the *only*  $\delta$ -sensitive clause of the Step-20 battery ( $\Theta\rho\Theta = \rho^{-1}$ ) is for  $\delta \neq \pi/2$  not violated but *not formulable* (a mark-compatible order-4 rotation exists iff  $\delta = \pi/2$ ) — it presupposes the bit instead of deriving it. The battery has teeth at every  $\delta$  (family A fails RP structurally

(4, 4, 0); the chirality- $\eta$  pinning persists (0, 8, 0)): RP separates the real-structure *families*, never the *sides*. Structure gained [C]: the group generated by the admissible OS reflections is  $V_4$  for every  $\delta < \pi/2$  and closes to  $D_4$  exactly at the clock point ( $\Theta_i \circ \Theta_1 = \text{the clock}$ ) — face #13 of the Step-2 equivalence web (12  $\rightarrow$  13). The free RP/ $\Theta$  battery is hereby the *eighth* side-blind test on the v512 scoreboard (7  $\rightarrow$  8); the honest [O] gap is the mark-decorated *state* class (defect/twist insertions, the interacting  $\mathcal{A}_{\text{hol}}$ ) — the mark-decorated half of that gap is since decided side-blind too (Step 26: the *ninth* test, 8  $\rightarrow$  9; the free-plus-twist class is exhausted, the residual gap narrows to the genuinely interacting  $\mathcal{A}_{\text{hol}}$ ) — the alignment bit remains genuine discrete input. No marker moves.  
[verification/v521\_seam\_bit\_rp\_blind.py]

**Step 23 — the clock is time-like, GSO is the gauge datum [E]/[C]/[O].** The  $\beta_1$  stage of the OS twistor bridge is executed as a *typed* result (verdict UNDECIDED per the frozen preregistration — the preregistered reading was the wrong operationalisation, and the module documents why, including its own first frozen run at 6/13 and a preregistration combinatorics error 30 vs 32). The mechanism: the one-step clock insertion  $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$  violates Hermiticity *exactly* (witness entry  $-i/(8 \sin(5\pi/16))$  against  $+i/(8 \sin(5\pi/16))$ ); the pairing is complex-symmetric — 745 matching, 96 anti-matching, 0 violations — and “OS-symmetric” is strictly weaker than Hermitian), so “positivity after gauge fixing” is not well-posed for the clock reading. The typing is a census: *all* 16 dihedral reflection axes of the NS seam circle invert the clock lift ( $RSR^{-1} = S^{-1}$ ), none commutes — the  $\mu_4$  clock *is* Woit’s euclidean rotation, and the gaugeable ( $\theta$ -compatible) part of its  $\mathbb{Z}_8$  Fock tower is exactly the 2-torsion  $\{1, (-1)^F\} = \text{the GSO/fermion-parity } \mathbb{Z}_2$  (the free-fermion shadow of the  $E_8$  glue:  $(E_8)_1 = \text{even NS of } SO(16)_1 + \text{R spinor}$ ). Under that corrected typing gauge-fixed RP *holds exactly*: (29, 0, 0) PD at  $N = 16$  (deg  $\leq 2$ , min eigenvalue  $1.78 \times 10^{-6}$ ), (8, 0, 0) PD on the complete  $N = 8$  half algebra; the site-cut defect survives gauge fixing ((7, 9, 6) — the bond placement is gauge-invariant information); family A stays indefinite ((17, 12, 0)); the Ramond control forces the NS/ $\mathbb{Z}_8$  tower. Kill test (2)’s free shadow does *not* fire, kill test (1) stays discharged also gauge-invariantly; both stay live on  $\mathcal{A}_{\text{hol}}$ , and the clock-equivariant statement is re-routed through  $\beta_2$  (OS quotient first, then the clock as reconstructed transfer operator — contract precision (iii)). No marker moves; WOIT.OS.TWISTOR.01 stays [O].  
[verification/v522\_woit\_beta1\_gso\_gauge.py]

**Step 24 — the  $w_m$  normalisation derived: the fixed-point factor is computed, not declared [E]/[C].** The named remaining target of Steps 17–21 — the *constructive* BCOV derivation of the completion normalisation  $1/\det_j$  — is now executed (verdict ERFOLG per the frozen preregistration), from three independent sources. *Route (i), Atiyah–Bott*: the equivariant mode ledger of the twistor fibre  $\mathbb{C}^2$  —  $c_n = \sum_{a+b=n} i^j(a-b)$  — equals the closed form  $1/((1-q i^j)(1-q i^{-j}))$  exactly in  $\mathbb{Q}(i)$  at every order  $n \leq 120$ ; the closed form is *regular* at  $q = 1$  (the twisted sectors carry no volume pole) with Abel value  $(1/2, 1/4, 1/2) = 1/\det_j$ ,  $\det_j = \det(1 - g^j) = (2, 4, 2)$ ; the  $(C, 2)$  Cesàro means converge in exact arithmetic; and the fixed-point factor *splits off exactly at every truncation level*  $d \in \{7, 12, 25\}$ , for the phased (skeleton) *and* the Step-21-forced unphased (completion) trace alike — the Abel-limit contact $_j = (Q^{(0)} - Q^{(j)})/\det_j$  is the Step-17 contact vector *computed*, not declared. *Route (ii), zeta/Quillen*: the spectral determinant of the  $g^j$ -twisted circle Laplacian is  $4 \sin^2(\pi j/4) = \det_j$  exactly (Lerch + reflection formula symbolically; Hurwitz certificates  $\sim 10^{-41}$ ); the Quillen split gives  $\det \Delta = \det_j^2$  with  $|\det \bar{\partial}| = \det_j$  and the real *positive* holomorphic section unique via the  $SU(2)$  conjugation pairing; and the  $\delta_{1f}$  modular block  $f_1 f_3$  at the  $(0, b)$  nodes carries the exact constant term  $1/\det_b$  (deviations  $\leq 5.2 \times 10^{-28}$ ) — the Step-19/21 transport machinery carries the fixed-point weight verbatim. *Route (iii), the consistency chain*: the derived weight reproduces the Step-17 chain number by number —  $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$ ,  $T_5 = 0$  at every scale,  $h_2 : h_1 = 4 : 3$ , the  $T_3$  budget  $32 - 8c$  forcing  $c = 4 = |\mu_4|$  uniquely, per-sector Okubo squares  $(18, 9, 18)\langle x, x \rangle^2$ , total  $45\langle x, x \rangle^2$ ,  $\Phi/\psi$  certificates  $\pm 64$ . The negative controls have teeth:  $1/\det^2$  leaves  $(T_5, T_3) = (2, 12)$  and contradicts the Quillen split;  $1/|1 - i^j|$  leaves  $\mathbb{Q}$  ( $w_1'' = 1 + \sqrt{2}$ ); the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor *passes* with the same derivation ( $\det = 4$ ,  $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ,  $c = 2 = |\mathbb{Z}_2|$ ); the  $SO(16)/D_8$  kill fires;  $\text{diag}(i, i)$  breaks the zeta = AB identity itself; and on  $\mathbb{C}^2/\mathbb{Z}_k$  for  $k = 3, 5$  the weights wander correctly

( $w_m = m(k-m)/2 = k h_m$  exactly in cyclotomic arithmetic) — only  $k = 4 = |\mu_4|$  carries the  $E_8$ -glue chain. Typed premises [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation *is* the definition of the equivariant character), TP-Q (the Quillen structure of BCOV  $F_1 = \text{Step-21 TP-4}$ ), TP-NUM (the unphased numerator = the Step-21 decision layer), TP-CH (the insertion channel computes the box-channel weight). Nothing in  $w_m$  is declared any more at this level; what stays [O] is the *global* BCOV integral on  $\mathbb{PT}/\mathbb{Z}_4$  beyond the fibre zero-mode factor (and the resolved-ALE route not built). No marker moves. [verification/v523\_celestial\_wp5e\_wm\_derived.py]

**Step 25 — the OS quotient made explicit: the clock gets its spectral calculus [E]/[C].**

The  $\beta_2$  stage of the OS twistor bridge is executed (verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii)).  $\mathcal{H}_{\text{phys}}$  is *explicit and nondegenerate*: at  $N = 16$  ( $\deg \leq 2$ ) the  $37 \times 37$  bond-cut OS Gram is exactly Hermitian and parity-block-diagonal with inertia  $(37, 0, 0)$  PD (min eigenvalue  $1.7801 \times 10^{-6}$  at 40 digits) and null space  $\{0\}$ ,  $\dim 37 = 29 \oplus 8$ ; at  $N = 8$  the *complete* half algebra gives  $(16, 0, 0)$  PD with  $\dim 16 = 8 \oplus 8 = 4^2$  — compact euclidean time reconstructs a *thermal* (KMS) representation, with the exact certificate  $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$ . The euclidean rotation becomes a Klein–Landau *local symmetric semigroup*:  $\tau_k$  is exactly Hermitian on every shrinking domain ( $k = 1..7$ ), the chain identities are exact, the vacuum is fixed, and the positivity pattern *is* the Step-20 site/bond dichotomy — even steps are PSD via the exact square identity  $T(2j) = A_j^* A_j$ , odd steps are indefinite with exactly zero one-particle diagonal (the one-step transfer is *not* positive: the free chirality datum, kill-test-3 shadow sharpened). The clock is the quarter turn  $T^{N/4}$ : positive self-adjoint on its domain  $((11, 0, 0)$ , min  $6.7 \times 10^{-5}$ ; compressed spectrum  $[0.0195, 1.6414]$  with vacuum eigenvalue exactly 1), with certified spectral projections ( $\sim 10^{-40}$ ) — exactly the calculus the non-Hermitian pre-quotient average of Step 23 could not have — and at  $N = 8$  the compressed clock spectrum is *exactly*  $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$  in both parity sectors: the silver axes of the Step-20  $\mu_4$  torsor return as the clock eigenvalue; the reconstructed rotation group  $U(s) = e^{isH}$  is unitary with the group law — per precision (iii) the [C]-operationalisation of “the clock acting unitarily”. The Step-23 non-Hermiticity is *resolved*: the census  $(745, 96, 0)$  is reproduced and every anti-matching entry is a wrap overlap (every overlap-free entry matches) — the pre-quotient failure was exactly the domain/wrap artifact. The pre-declared KMS deviation arrives with exactly the declared witnesses: no contraction on the compact circle (the edge mode expands by  $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$  exactly;  $\det(G - \tau_4) < 0$ ). GSO/ $\Theta$ : the grading survives, the *perpendicular* torsor mirror descends anti-unitarily with  $\Theta_{\text{phys}}^2 = +1$  on every sector (the quotient is Kramers-free), and  $\theta_{\text{cut}} \circ \theta_{\perp} = \alpha_{N/2}$  exactly (the torsor closes onto the deck). Controls: the site cut is indefinite (the contract kill branch fires for the site placement), family A admits no quotient, the anti-chiral state flips to  $(8, 8, 0)$  — quotient existence itself selects the chiral orientation. Kill tests (1)/(2) strengthened, (3) shadow sharpened, none fires; all seven stay live on  $\mathcal{A}_{\text{hol}}$ ;  $\beta_3$  is next. Transparency: run 1 scored 18/20 — a sympy simplify failure on a *true*  $\pi/16$  product formula, fixed by a Laurent-polynomial certificate; no criterion changed. No marker moves; WOIT.OS.TWISTOR.01 stays [O]. [verification/v524\_woit\_beta2\_os\_quotient.py]

**Step 26 — the twist-state kill: the ninth side-blind test, and the free-plus-twist class is exhausted [E]/[O].**

Step 22 left one named door: marks decorating the *state* (defect/twist insertions). That door is now decided *negative*, exactly as preregistered (KILL) — the *ninth* side-blind test on the v512 scoreboard ( $8 \rightarrow 9$ ). Every well-defined mark-decorated state functional in the free-plus-twist class is RP-blind on the always-existing cuts along the whole  $M(\delta)$  ladder: the  $\sigma$ -gauge arc decoration satisfies  $\sigma \circ r = \sigma$  on the swap axes, so the twisted Gram is a diagonal unitary congruence — spectra *identical* to 40 digits; the Kadanoff–Ceva 4-twist insertion  $\omega(D \cdot)/\omega(D)$  is well-defined on the  $N = 32$  ladder ( $\omega(D)$  real nonzero; the  $N = 16$  odd- $m$  degeneration is the checkerboard artifact) and its RP spectra are the *first* free-class data of the programme that depend on  $\delta$  at all (pairwise up to 1.05) — yet the *inertia* stays  $(16, 0, 0)$  PD with the same  $\eta = +i$  for every member  $m = 1..7$ : the decoration sees  $\delta$ , the positivity does not; the  $\mu_4$  defect at  $\beta = \pi/2$  is pure plane gauge, and the genuine Bogoliubov defect at  $\beta = \pi/4$  — where  $\Theta$ -compatibility selects exactly 2 of 16 sign patterns (the crossed pattern forced by the NS wrap signs) and the winner is genuinely non-monomial — is still PD with spectra again identical. The structure theorem behind

the blindness is exact: the defect planes never straddle a cut bond, so every  $\Theta$ -compatible *quasi-free* mark decoration is RP-spectrally *invisible* (orthogonal congruence). The sub-results sharpen the verdict: the defect energy  $E_{\text{def}} = 1.2752872$  is exactly  $\delta$ -independent (spread  $4 \times 10^{-40}$ ); the 4-twist Casimir  $\ln |\omega(D)|$  *measures*  $\delta$  smoothly but is exactly mirror-symmetric under  $m \leftrightarrow 8 - m$  — the  $\delta \leftrightarrow \pi - \delta$  equivariance makes  $\pi/2$  the symmetric point of every invariant readout: a family gauge, never a side selector; the twisted parity  $\langle \Gamma \rangle_{\text{tw}} = +1$  for every member. The harvest: two genuinely twist-visible structures — the  $\eta$  flip  $+i \rightarrow -i$  on the mark-fixing axes ( $\sigma \circ r = -\sigma$ ) and the twist frustration of the mark-fixing mirror  $((4, 4, 0)$  indefinite for both  $\eta$ : the lattice shadow of the Ising twist field’s mirror-oddness) — both live *only* on the fixing axes existing iff  $\delta = \pi/2$ : they presuppose the bit (the **v512** facet class). Controls: the twist-free limit reproduces Steps 20/22 exactly; the site cut keeps failing; the **v512** counterwitness is arc-equivalent at  $N = 16$  to the blind member  $\pi/4$  — no state-level selector can exclude it; a single twist has  $\omega(D) = 0$  identically (twists exist only in mark pairs). The free-plus-twist class — everything Wick-computable — is hereby *exhausted*: what could still see the bit is only a genuinely *interacting*  $\mathcal{A}_{\text{hol}}$  whose OS data are not Pfaffian-reducible to the chiral vacuum — and that door is since probed by Step 30 (the tenth side-blind test  $9 \rightarrow 10$ , with Kill-Test 2 firing at toy level under a typed fence). The alignment bit remains genuine discrete input. No marker moves. [[verification/v525\\_seam\\_bit\\_twist\\_blind.py](#)]

**Step 27 — the thermal seam: KMS closes onto the Nariai/BH chain as the third leg of  $c_3$  [E]/[C].** The free OS quotient of Step 25 is not merely thermal in dimension  $(4^2)$ : its temperature is *measured*, not assumed. The circle kernel  $C(d) = (2/N)/\sin(\pi d/N)$  admits exactly one finite thermal-mode representation with detailed balance — at  $N = 8$  one real frequency  $\omega = \text{arcsinh}(2^{-3/4})$  with weight ratio  $\kappa = q^{-8}$  exactly ( $\beta = 8 = N$ ; Q-polynomial identity  $Q^2 - (2 + \sqrt{2})Q + 1 = 0$ ), at  $N = 16$  both mode pairs real with pairing exponent  $(N/2) + 1 = 9$  ( $\beta = 16 = N$ ), and wrong candidates  $\{N/2, N/4, 2N\}$  are excluded by quantified gaps. In angle units (one clock step =  $2\pi/N$ , the **v519** pin) one obtains  $\beta_{\text{angle}} = 2\pi$  exact, hence  $T_{\text{seam}} = 1/(2\pi) = 4c_3$  —  $N$ -independent, and exactly the Bisognano–Wichmann/Hawking modular normalisation that **v208** carries into the horizon sector. The modular Hamiltonian is closed ( $\nu_1 = \cos(\pi/24)$ ,  $\nu_2 = \cos(7\pi/24)$ ;  $\varepsilon_i = 2 \ln \cot(\theta_i/2)$ ; Tomita KMS- $\beta = 1$  on all 256 monomial pairs). The anti-numerology audit selects the class-1 identity  $r = 1$ : temperature joins geometry and anomaly as the *third leg* of  $c_3$ . The Hawking/Nariai chain then reads in the same normalisation:  $T_H = c_3/M = 1/(8\pi M)$  (the axiom *is* the Hawking coefficient, now from the seam KMS), SdS  $1/(4\pi) = 2c_3$ , Nariai  $T_N = 4c_3\sqrt{\Lambda}$ , and the **v104** cross-link  $2H/T_N = 4\pi = 1/(2c_3)$  exact. Entropy confirms the chiral  $c = 1/2$  log-law ( $\Delta S/\ln 2 \rightarrow 1/6$ ) but the **v129** horizon fractions  $\{1/3, 2/3\}$  and the **v190** floor do *not* transfer — the temperature bridge closes, the entropy-fraction bridge honestly does not. Typed non-gating: the naive clock reading  $\{1, \sqrt{2} - 1\} = e^{-\Delta K}$  fails; the exact replacement is  $\sqrt{2} - 1 = C(3)/C(1) = 1/(2 \cosh 2\omega - 1)$  — the silver eigenvalue is a thermal kernel ratio (Step 28). Non-compact control: the Stieltjes half-line has  $\kappa = 0$ ,  $\beta = \infty$ ,  $T = 0$  — the temperature hangs exactly on compactness of the euclidean circle. [C]-fence: the reading “seam euclidean circle = thermal circle of the reconstructed horizon dynamics”. No marker moves. [[verification/v526\\_seam\\_thermal\\_kms\\_nariai\\_bridge.py](#)]

**Step 28 — the silver eigenvalue explained and demystified [E].** The compressed clock spectrum  $\{1, \sqrt{2} - 1\}$  of Step 25 is now closed-form: at  $N = 8$  the Hankel pencil is diagonal,  $\tan(\pi/8) = C(3)/C(1)$ , the doubling is exact fermionic second quantisation, the geometric carrier is the silver-midpoint axis of the  $\mu_4$  torsor, and the gap is  $\text{arcsinh}(1) = \text{gd}^{-1}(\pi/4)$ .  $N$ -scaling ( $N = 8..24$ ) shows a pure  $N = 8/16$  Majorana *lattice* datum — not a continuum limit; a positive  $\mu_4$  clock exists only for  $N \equiv 0 \pmod{8}$ ; the golden family appears as the  $N = 10$  control. The connection verdict is an honest KILL in the **v100/v513** style: a preregistered 11466-candidate census over the 16 frozen registry numbers yields 0 hits above null expectation, with golden/plastic placebos comparable — no readout connection. Cross-link to Step 27:  $\sqrt{2} - 1$  is the thermal kernel ratio. Double verdict; no marker moves. [[verification/v527\\_seam\\_clock\\_silver\\_spectrum.py](#)]

**Step 29 — the bit as a twist-class choice: the physical definition [E]/[C].** Step 26 harvested two bit-presupposing  $\pi/2$  structures (the  $\eta$  flip and the twist frustration). Both are now exact



*equivalences* to the bit, both directions each. Facet #14:  $\eta$ -flip  $\Leftrightarrow \delta = \pi/2$  ( $\sigma \circ r = -\sigma$  on all four fixing axes; off  $\pi/2$  on every axis  $\sigma \circ r = +\sigma$ ; the cardinality lemma  $\sigma \circ f = -\sigma \Rightarrow |A_+| = |A_-| \Rightarrow m = 4$ ;  $g$ -class census over all 224 axes: flip class exactly at  $\{15, 31\}$  for  $m = 4$ ). Facet #15: twist-frustration  $\Leftrightarrow \delta = \pi/2$  ( $\theta$  exchanges  $D_{\text{short}} \leftrightarrow D_{\text{long}}$ ; Gram indefinite  $(4, 4, 0)/(8, 8, 0)$  exactly at  $\pi/2$ ; off reducible and PD; same cardinality lemma). The order parameter  $O = \text{flip-axis fraction}$  equals  $1/2$  at  $m = 4$  and 0 everywhere else, with class alternation  $(-, +, -, +)$  the nontrivial  $\mathbb{Z}_2$  character of the  $\mathbb{Z}_8$  clock-tower shadow; gauge-robust under every v525 freedom. Unification:  $\eta$ -flip = frustration = flag-transitivity are *one*  $\mathbb{Z}_2$  invariant (same arc system); the v510 Kramers class is another (constant on the family) — Kramers = POSITION half (derived), twist class = MODULUS half (the input). The v512 web grows  $13 \rightarrow 15$  facets. [C] measurement sketch (principle only): defect transport around the axis torsor accumulates the  $\eta$  holonomy = a relative  $\pi$  phase iff the bit is set — in principle interferometrically readable. Honest fence: the bit remains formal input; a definition replaces no derivation. No marker moves. [verification/v528\_seam\_bit\_twist\_class\_definition.py]

### Step 30 — the interacting FK toy: the first firing Kill-Test-2 shadow [E]/[C]/[O].

The door left open by Steps 22/26 — a genuinely interacting  $\mathcal{A}_{\text{hol}}$  — is probed on the minimal Fidkowski–Kitaev quartic on the 16-Majorana NS seam circle (256-dim, exact). Triple verdict. F1 = KILL (the most important part):  $\Theta$  exists exactly on the interacting model ( $U_r^2 = 2^8$ ,  $\Theta^2 = +1$ , clock-tower inversion  $\text{conj}_V \cdot V = 4096$ ,  $[\Theta, H_g] = 0$ ,  $\Theta\Omega_g = \Omega_g$ ) — Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: reflection positivity breaks in the interacting ground state for every  $g > 0$  — inertia ladder  $(37, 0, 0) \rightarrow (33, 4, 0) \rightarrow (31, 6, 0) \rightarrow (29, 8, 0)$  (most negative eigenvalue  $-4.5 \times 10^{-2}$ ); the Gibbs state at  $\beta = 2$  carries the same ladder; the mechanism is interference (neither crossing nor half-local terms alone break); the STRADDLE LAW holds on all 24 entries — RP fails exactly on quartet-straddled cuts and stays PD on quartet-avoiding ones. Mandatory fence: one toy, one interaction class, [C] flat-band parent — this *narrows* the Woit route (every  $\mathcal{A}_{\text{hol}}$  must protect RP against exactly this mechanism; the straddle law is a concrete hurdle for  $\beta_3/\gamma$ ), it does *not* close it; WOIT.OS.TWISTOR.01 stays [O]. F2 = KILL: the *tenth* side-blind test on the v512 scoreboard ( $9 \rightarrow 10$ ), the first with genuinely non-Wick-computable dynamics — the interaction sees  $\delta$  massively (spectral spread 1.32), but the mark-axis mirror  $\Theta_{15}$  maps  $H(m) \rightarrow H(8 - m)$  exactly; all side data are mirror-equal. F3 = ERFOLG: grading and the nonsplit  $\mathbb{Z}_8$  tower survive ( $\tilde{U}^2 = 256\Gamma = (-1)^F$ ,  $V^2 = 256\tilde{U}$ ); the FK fixed-point anchor holds; the v524 OS quotient survives  $g > 0$  constructively on the  $\delta = \pi/2$  mark-anchored model. No marker moves. [verification/v529\_seam\_interacting\_toy\_fk.py]

**What remains, honestly (WP5e proper [O]).** The single open step is the *global* BCOV/Kodaira–Spencer quantisation on  $\mathbb{P}\mathbb{T}/\mathbb{Z}_4$ : the boundary partition function  $E_4/\eta^8$  *derived from the twistor side*, including the  $q^{-1/3}$  prefactor whose CFT side Step 9 has pinned and whose equivariant skeleton Step 11 has verified — the exchange sub-branch is closed by Step 12, the full-tensor ledger is executed by Step 14, the level-from-flux dial is executed by Step 13, the bulk-axion slot is built by Step 15, all three back-reaction milestones it preregistered are executed by Steps 16–18 (the Step-15 fence M1–M3 is *fully worked off*), the  $\delta_1$  chain is *decided* by Step 19 (kill under the derived measure), and the measure question is *decided at probe level* by Step 21 (the declared completion reading wins: single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4; the derived chiral measure was never a well-defined nonvanishing moduli integral), and the constructive BCOV derivation of the  $w_m$  normalisation itself is *executed* by Step 24 ( $1/\det_j$  computed from three independent sources; nothing in  $w_m$  is declared any more at that level), so the named remaining target narrows to the *global BCOV integral on  $\mathbb{P}\mathbb{T}/\mathbb{Z}_4$  beyond the fibre zero-mode factor*; the cubic  $d$ -channel stays not needed and its physical-justification question ( $c_d = 32 \times 60$ ; the  $1920 = |W(D_5)|$  reading look-elsewhere-loaded and convention-contingent, v513) stays dissolved (Steps 19 and 21 do not revive it) — together with the continuum uplift of the Step-10 lattice witnesses (Xu’s theorem for the abstract orbifold statement; the concrete seam quotient net and the interacting condensed  $(E_8)_1$  net are Costello–Li territory). The route does *not* close SEAM.EQUIV.01; it *is* the keystone’s second, quantitative route, named in its own right as SEAM.EQUIV.TWISTOR.01 ([O]; the conditional MMST route is SEAM.EQUIV.MMST.01, closed

modulo cited theorems; the parent closes if either route closes and stays [O] as an unconditional claim) — the same two-step shape as MMST (limit + maximal ideal) with the ideal quantitatively fixed, the deleting operator explicit, the limit state constructed, the index chain measured, the KLM triple witnessed, both faces of the inflow anchored, the collapse confirmed full-tensorially with one cubic door open, one level a theorem with its value pinned by the sector counter, the bulk-axion slot a construction with  $\tilde{\lambda} = 6$  triply pinned, the back-reacted  $\Omega_N$  closed-form with  $(2\pi i)^2$ -integral lockstep periods and the source charge  $4 = |\mu_4|$  forced, the twisted KS measure (on the declared completion reading) landing on the unique quartic-free weighting, the  $a_0$  uplift coupled to the centre count on four scales, the  $\delta_1$  chain decided — kill under the derived measure — the measure question decided at probe level for the declared reading (Step 21) and the  $w_m$  normalisation itself derived constructively (Step 24) — while the global quantisation (now narrowed to the global BCOV integral beyond the fibre zero-mode factor) stays open. SEAM.EQUIV.01 stays [O] throughout; no marker moves.

**The Woit bridge: chiral Wick rotation and the real structure [O].** One *external* programme is close enough in shape to deserve a named paragraph — named precisely so that proximity is not mistaken for confirmation. Woit’s *Euclidean Twistor Unification* (arXiv:2104.05099) starts from the observation that the usual Wick rotation is problematic for *purely chiral* theories (the Euclidean continuation of a chiral fermion is not a chiral fermion;  $SO(4) \cong SU(2) \times SU(2)$  splits what  $SO^+(3, 1)$  holds together), and proposes the counter-move: formulate the theory *Euclidean-holomorphically on projective twistor space*  $\mathbb{PT}$ , and reconstruct the Lorentzian theory from a conjugation structure / boundary values — the real form is *recovered*, not assumed. TFPT’s celestial/twistor route has, independently, assembled the very ingredients such a reconstruction needs: reflection positivity of the seam collar (v379; the Osterwalder–Schrader axiom, machine-witnessed), the OS reconstruction step itself (the cited AMT/OS selector in FORM.SEAM.MMST.01), the order-4 clock  $\rho = \text{diag}(i, 1)$  acting on the twistor sphere (v492), and the seam reflection (Step 20 makes the referent precise: the seam-circle *reflection*, not the deck/covering involution — the deck, whose freeness is topology v510, carries the  $(-1)^F$  Kramers class instead). What is *missing* is stated exactly: a *global anti-linear map*  $\Theta : \mathcal{A}(\mathbb{PT}/\Gamma) \rightarrow \mathcal{A}(\mathbb{PT}/\Gamma)$  with  $\Theta^2 = 1$  and  $\Theta\rho\Theta = \rho^{-1}$  on the *interacting* open+closed twistorial algebra, together with reflection positivity of the interacting BCOV+SDYM functional — the two inputs of the new central contract WOIT.OS.TWISTOR.01 in `tfpt_research_contracts`. The  $\alpha$ -stage status (Step 20, v519): *the missing global object now exists at the free level* —  $\Theta$  with  $\Theta^2 = +1$  and  $\Theta\rho\Theta = \rho^{-1}$  on all four levels (sphere,  $\mathbb{C}^4$ ,  $\mathbb{Z}_8$  spin,  $\text{Cl}(16)$  Fock), with free RP on the seam system selecting the same family; Woit’s two inequivalent real structures ( $\rho_{\text{tw}}$  and  $\sigma_{\text{std}}$ ) are exactly the two families of that classification, in *different* contract slots. The  $\beta_1$ -stage status (Step 23, v522) sharpens the dictionary once more: the  $\mu_4$  clock is *Woit’s euclidean rotation itself* (time-like: every seam mirror axis inverts it), the only gaugeable part of its tower is the GSO/fermion-parity  $\mathbb{Z}_2$ , and gauge-fixed RP holds under that typing — kill test (2)’s free shadow does not fire. The  $\beta_2$ -stage status (Step 25, v524) delivers the first *reconstructed* objects: the OS quotient of the free system is explicit —  $(\mathcal{H}_{\text{phys}}, \Omega)$  positive definite at both levels, the euclidean rotation a positive transfer step with spectral calculus and a reconstructed unitary rotation group (exact clock spectrum  $\{1, \sqrt{2} - 1\}$  at  $N = 8$ ), the compact-circle reconstruction thermal (KMS) with exactly the pre-declared silver witnesses. *The interacting statement stays [O]* — kill tests 1 and 2 are discharged only at the free level, and until  $\Theta + \text{RP}$  exist on  $\mathcal{A}_{\text{hol}}$ , the Woit connection remains a strong analogy, not a mathematical integration; nothing in this paragraph is evidence for either programme, and no marker moves.

## 18 Honest status: bootstrap, not creation from nothing

- **What is genuinely shown [E]:**  $g_{\text{car}} = 5$  is an *overdetermined* fixed point of the  $E_8$  closure (two independent Lie-theory conditions); the integer “8” in  $c_3$  is the carrier Coxeter =  $E_8$  rank; the discrete data closes a loop. The earlier landscape scan confirms  $g_{\text{car}} = 5$  is the *unique* value at which all hierarchies/family/gauge close — so the fixed point is unique, not



one of many. **This uniqueness is now machine-formalised [E] in Lean 4** (Theorem A, `lean4-carrier-rigidity`): the division-free Pascal condition  $2^g = g^2 + g + 2$  has the *unique* solution  $g = 5$  (`carrier_rank_pascal_unique`; growth lemma for  $g \geq 6$ ;  $N_{\text{fam}} = (2^4 - 1)/5 = 3$ ,  $g + N_{\text{fam}} = 8$ ), audited to use only the standard kernel axioms. The 2026-06-11 hardening round adds two formal modules: `AnchorLadder.lean` (the anchor power-sum law  $p_n(a) = 2 + 2^n$  for *all*  $n$ , the ladder atoms (3, 4, 6, 10), 240/8/248, and the binary ladder  $p_{n+1} - p_n = 2^n$  as general theorems) and `PascalLadder.lean` (the odd-midpoint ladder identity  $\sum_{k \leq K} \binom{2K+1}{k} = 2^{2K}$  for *all*  $K$  via Mathlib, plus the kernel-decidable bounded uniqueness of the  $\bar{K}=2$  carrier case) — the solves-side of the Pascal Ladder Theorem ([`verification/v108_pascal_ladder.py`]) is thereby [E].

- **What this is *not*:** a derivation from zero inputs. A self-consistency loop is consistent *at* a value; uniqueness is what makes it predictive (and that is supplied by the landscape scan). The continuous  $\pi$  is not produced by the loop — it is the Möbius/boundary geometry, the one irreducible primitive.
- **Conceptual gain:** the theory’s foundation tightens from “two free axioms” to “*the*  $E_8$ -closure fixed point + the geometric  $\pi$ .” That is the Möbius self-consistency you intuited: the seed is the start of the cascade, and the cascade returns the seed’s discrete content.
- **The current reduction state (2026-07-22; text consolidation, markers unchanged):** the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *one discrete symmetry-lift bit* (flag transitivity  $\iff \tau = i$ ; equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v499/v506/v507/v510/v512), and  $\pi$  — with the [C] identifications R1/R2 of v499 as the remaining conditional layer. AX.P2.01 stays a declared axiom and QGEO.REALIZE.01 stays [C]/[O]: this line describes the *reduction state*, not an abolition.

## Part IV

Open items and the  $E_8$ -slice scan

## 19 The honest open-items list

This is the complete current ledger of what is *not* fully closed, graded by how hard the gap is.

| #                                                                                     | Open item                                                  | Grade   | Hardening path / what closes it                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------|------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>1. SM core — finite computations, no new physics</i>                               |                                                            |         |                                                                                                                                                                                                                   |
| 1a                                                                                    | absolute quark $c$ -scale (ratios closed)                  | [O]     | quark <i>ratios</i> are integer Plücker readouts on the derived selector stratum (v49/v69/v71); only the <i>absolute</i> $U_f^*$ normalisation remains — an anchor                                                |
| 1b                                                                                    | $\theta_{12}$ solar angle                                  | [E]     | <b>texture closed (below):</b> explicit $\mu\tau$ -symmetric texture verified; only $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} \approx c_3$ (0.23%) stays conditional                                       |
| <i>2. Scheme layer — standard RG, no new input</i>                                    |                                                            |         |                                                                                                                                                                                                                   |
| 2a                                                                                    | $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$                 | [C]     | RG threshold matching $R_{\text{SM}}; m_H$ on the closed UV quartic $1/16\pi^2$                                                                                                                                   |
| 2b                                                                                    | hadron pole spectrum (singlet)                             | [C]     | singlet algebra $\mathcal{A}_{\text{sing}}$ explicit; dynamics already closed (gap $6 \log \frac{3}{2}$ )                                                                                                         |
| <i>3. Conditional — named analytic hypotheses</i>                                     |                                                            |         |                                                                                                                                                                                                                   |
| 3a                                                                                    | gravity: local eq. parameter-free; ambient redundancy      | [E]/[C] | full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ (v358/v359); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure a [C] redundancy (v369) |
| 3b                                                                                    | strong CP / QFT closure                                    | [C]     | reflection positivity (RP) is the one remaining input                                                                                                                                                             |
| 3c                                                                                    | $\Lambda$ branch label (rec vs car)                        | [C]     | one discrete choice; typed via $(8\pi)^2$ , value not fixed                                                                                                                                                       |
| <i>4. Axioms — not eliminable, only hardenable</i>                                    |                                                            |         |                                                                                                                                                                                                                   |
| 4                                                                                     | P1 ( $c_3$ ), P2 ( $g_{\text{car}}=5$ )                    | [O]     | more Lean; P2 algebra already machine-verified; remain inputs                                                                                                                                                     |
| <i>5. Frontier — typed transfer targets, never compiler powers (see frontier doc)</i> |                                                            |         |                                                                                                                                                                                                                   |
| 5                                                                                     | $\eta_B, m_p/m_e$ , Koide, dark matter, ambient QG measure | [C]     | transfers now runnable solvers with kill tests (v371–v375); ambient measure a [C] redundancy (v369); not forced (scan results below)                                                                              |
| <i>6. Cosmology late-time pipeline — not simplified by the compiler</i>               |                                                            |         |                                                                                                                                                                                                                   |
| 6                                                                                     | $C_\ell$ transfer, $a_{\ell m}$ seam, baryogenesis         | [C]     | frontier-doc machinery; $a_{\ell m}$ “good CMB world, not yet this CMB world” (SMICA test)                                                                                                                        |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

**Net.** Items 1 are finite arithmetic; 2 are standard RG; 3 hinge on named hypotheses (RP, the  $\Lambda$  branch, the UV completion); 4 are the two irreducible axioms; 5–6 are honestly open. The inflation sector (the only cosmology piece the compiler *does* sharpen) is now a parameter-free prediction ( $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ , document 2).

#### Item 1b closed: the explicit solar-angle Majorana texture [E]

The last open SM angle is now realised by an *explicit* texture, not a fitted number. Take the  $\mu\tau$ -symmetric Majorana matrix (tri-bimaximal at  $\varepsilon=0$ )

$$M_\nu(\varepsilon) = \begin{pmatrix} -\eta(\varepsilon) & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \eta(\varepsilon) = \frac{2\sqrt{2}}{\tan 2\theta_{12}} - 1, \quad \sin^2 \theta_{12} = \frac{1}{3}(1 - 2\varepsilon). \quad (1)$$

$\mu\tau$ -symmetry forces  $\theta_{23} = 45^\circ$  and  $\theta_{13} = 0$  *exactly*; diagonalisation at the TFPT solar deviation  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$

gives

$$\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747, \quad \theta_{23} = 45^\circ, \quad \theta_{13} = 0 \quad [\mathbf{E}] \quad (2)$$

**Honest residual** ([verification/v9\_neutrino\_texture.py]). The texture and the angle are exact. The seam identification is  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = \frac{3}{4}\left(\frac{1}{6\pi} + \frac{3}{256\pi^4}\right) = c_3 + \frac{9}{1024\pi^4}$ : the *leading geometric term is exactly*  $c_3 = \frac{1}{8\pi}$ , and the 0.23% residual is purely the  $\varphi_0^{\text{ret}}$  seed tail (document 2). So  $\varepsilon = c_3$  holds at leading order; the full  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  is what enters the texture. The reactor angle  $\theta_{13}$  ( $\sin^2 \theta_{13} = e^{-5/6}\varphi_0^{\text{ret}}$ , document 2) enters through a separate  $\mu\tau$ -breaking term and is not part of this minimal solar texture. **Scope:** this realises the TFPT solar shift in the  $\mu\tau$ -symmetric *limit*; it is *not* the full PMNS matrix. The  $\theta_{23}=45^\circ$  value is the symmetric-limit centre, *not* an octant prediction — current global fits (NuFIT 6.0) leave the octant unresolved — and the reactor angle plus the combined matrix require the separately stated  $\mu\tau$ -breaking channel.

## 20 The $E_8$ -slice scan: results (honest hit / miss)

All seven maximal slices of 248 were scanned for the unassigned quantities. The rule was a meaningful (low-complexity) compiler/slice expression matching to  $< 0.5\%$ ; matches that need a free fitted factor are reported as *misses*.

| Slice / target                       | Result                                                                                                                                                                                             | Verdict                   |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| $A_4 \times A_4 = SU(5)^2$           | $\sin^2 \theta_W _{\text{GUT}} = 3/8$ sits in the $SU(5)$ hypercharge embedding; the 12 coset states are the proton-decay $X, Y$ bosons                                                            | $[\mathbf{E}]$ (known)    |
| $F_4 \times G_2$ / Koide             | Koide $Q = 2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ (sheet over families); $J_3(\mathbb{O})$ has $\dim 27=26+1$ , $3 \times 3$ octonionic “generations” on the diagonal                                | $[\mathbf{C}]$ structural |
| $D_8 = SO(16)$                       | $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ — <i>same</i> spinor as $D_5 \times A_3$ ; $\nu_R$ (seesaw) lives in the 16; <b>no new dark-matter state</b> (only the 4th $\mu_4$ direction) | $[\mathbf{E}]$            |
| $A_8 = SU(9)$ , $9=3 \times 3$       | $\Lambda^3(3, 3) = (10, 1) \oplus (8, 8) \oplus (1, 10)$ : the $(8, 8)=64$ dominates — naive family $\times$ colour <b>fails</b>                                                                   | miss                      |
| $A_8 \supset SU(5) \times SU(4)$     | $\Lambda^3(5+4)$ contains the SM <b>10</b> : a genuine chiral-family-unification <i>lead</i> , but not TFPT-forced                                                                                 | lead                      |
| $E_6 \times A_2$ / Jarlskog          | cross term $2(\mathbf{1}^\top Ra)N_{\text{fam}} = 162 = 248 - 78 - 8$ exact; $J \sim \lambda_Y^6$ only to right order, no clean hit                                                                | weak                      |
| $E_8$ theta shells                   | root shells $240, 2160, 6720, \dots$ ; $31=2^{q_{\text{car}}}-1$ is <i>not</i> a clean shell ratio                                                                                                 | inconcl.                  |
| $m_p/m_e = 1836.15$                  | no clean $E_8$ /compiler expression (a QCD-confinement ratio); apparent hits need a free factor                                                                                                    | <b>miss</b>               |
| $\eta_B \approx 6.1 \times 10^{-10}$ | sits <i>between</i> $c_3^6=4 \times 10^{-9}$ and $c_3^7=1.6 \times 10^{-10}$ ; right order, no integer power                                                                                       | order only                |

Marker key:  $[\mathbf{E}]$  exact / machine-proven ·  $[\mathbf{C}]$  conditional (holds under named hypotheses) ·  $[\mathbf{O}]$  open / axiom ·  $[\mathbf{X}]$  falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

### Genuine new readings the scan produced

- **Koide** =  $|\mathbb{Z}_2|/N_{\text{fam}}$ . The empirical Koide value  $2/3$  is exactly the sheet over the family count — a clean (if simple) compiler reading, promoting Koide from “near  $2/3$ ” to “democratic  $|\mathbb{Z}_2|/N_{\text{fam}}$ .”
- **$E_8$  as the optimal 8-bit code.**  $E_8$  is the optimal 8D lattice (kissing number 240). In byte framing  $256-248 = 8 = \text{rank } E_8$  and  $256-240 = 16 = \dim S^+$  — a clean information-theoretic reading of the two load-bearing  $E_8$  numbers.
- **Dark matter = no new state.** The  $SO(16)$  scan shows the 128 is the *same* matter spinor; there is no spare  $E_8$  singlet to be a WIMP. Any DM candidate must be the determinant-line axion (frontier doc), not a new  $E_8$  rep — a useful *negative*.
- $SU(9) \supset SU(5) \times SU(4)$  **chiral lead.**  $\Lambda^3$  puts the SM **10** in the antisymmetric; the cleanest unexploited route to a family-unification reading, flagged for future work.

**Honest negatives**

$m_p/m_e$  has *no* clean  $E_8$ /compiler derivation (it is a confinement ratio — the apparent “hits” all need a free fitted factor);  $\eta_B$  matches only at the order-of-magnitude level ( $\sim c_3^{6.x}$ ); the  $\Lambda$  rec/car branch is *not* selected by the theta series; the Jarlskog invariant is right only to order  $\lambda_Y^6$ . These are reported as open, not forced.

**21 Following the  $SU(9)$  lead: cross-slice consistency**

The one scan lead worth following is  $A_8 = SU(9) \supset SU(5) \times SU(4)$ . Splitting  $9 = (5, 1) \oplus (1, 4)$  the  $E_8$  branching  $248 = 80 \oplus 84 \oplus \overline{84}$  decomposes *exactly* as:

| piece              | $SU(5) \times SU(4)$ content                                                                 | reading                                               |
|--------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 80 (adj)           | $(24, 1) + (1, 15) + (5, \overline{4}) + (\overline{5}, 4) + (1, 1)$                         | $SU(5)_{\text{GUT}} + SU(4)_{\text{fam}}$ gauge       |
| $84 = \Lambda^3 9$ | $(10, 1) + (10, 4) + (5, 6) + (1, 4)$                                                        | the SM <b>10</b> as family-singlet + family- <b>4</b> |
| $\overline{84}$    | $(\overline{10}, 1) + (\overline{10}, \overline{4}) + (\overline{5}, 6) + (1, \overline{4})$ | the conjugate matter                                  |

(Dimension checks:  $24+15+20+20+1 = 80$ ;  $10+40+30+4 = 84$ ; total 248.)

**The genuine content:  $A_8$  and  $D_5 \times A_3$  agree on the family [E]**

The  $SU(4)$  here is *the same*  $A_3$  family group as in the construction slice  $D_5 \times A_3$  — the four punctures  $\mu_4$  of  $\mathbb{P}^1 \setminus \mu_4$ . The  $SU(5)$  is the Georgi–Glashow GUT sitting inside the carrier  $SO(10) = D_5$  ( $SU(5) \subset SO(10)$ ). So the two maximal slices are *consistent*: both read the family as  $SU(4)=A_3$ , and  $A_8$  additionally exposes the  $SU(5)$  matter content — the SM **10** appears as  $(10, 1) \oplus (10, 4)$ , i.e. a family-singlet **10** plus a family-**4**, where the **4** of  $A_3$  is exactly TFPT’s “ $4 = N_{\text{fam}} + 1$ ” (three families + the  $\mu_4$  direction).

**Honest limit: chirality is still the index, not  $E_8$** 

$84 \oplus \overline{84}$  is *vectorlike* ( $10 \oplus \overline{10}$  etc.):  $E_8$  being non-chiral, the slice cannot by itself produce three *chiral* generations. The chirality is the boundary index  $\text{Ind}(D_{\text{fam}}^+) = 3$  — the *same* input TFPT already uses. So  $SU(9)$  is a consistent re-reading (matter in  $SU(5)$ , family in  $A_3$ ), not an independent derivation of  $N_{\text{fam}} = 3$ . Each  $SU(5)$  pair (**10** + **5**) is anomaly-free, so the embedding is clean. [C]

**Appendix: computational verification**

The audit/bridge identities of this document (marked [E], [E]) are re-derived from  $\{c_3, g_{\text{car}}\}$  and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the  $\mu_4$  glue  $E_8 = (D_5 \oplus A_3) + \mu_4$  (load-bearing).
- `v4_flavor_matrix.py` — the spectral selector  $\det R = h(D_5) = 8$ , minors  $\{2, 3, 5\}$ .
- `v5_e8_cascade.py` — the cascade  $D_n = 60 - 2n$ : endpoints, exponent rungs, IR tail, variance.
- `v6_bootstrap.py` — the self-consistency loop:  $\mu^2 - 5\mu + 4 = 0$ ,  $g_{\text{car}} = 5$  three ways,  $8 = \text{rank } E_8$ .

The  $E_8$ -slice/atlas readings are explicitly [O] (audit-level), not part of the pass/fail suite. Run `python verification/run_all.py`; full map in `verification/README.md`.